

ECE 3337: Signals & Systems Analysis

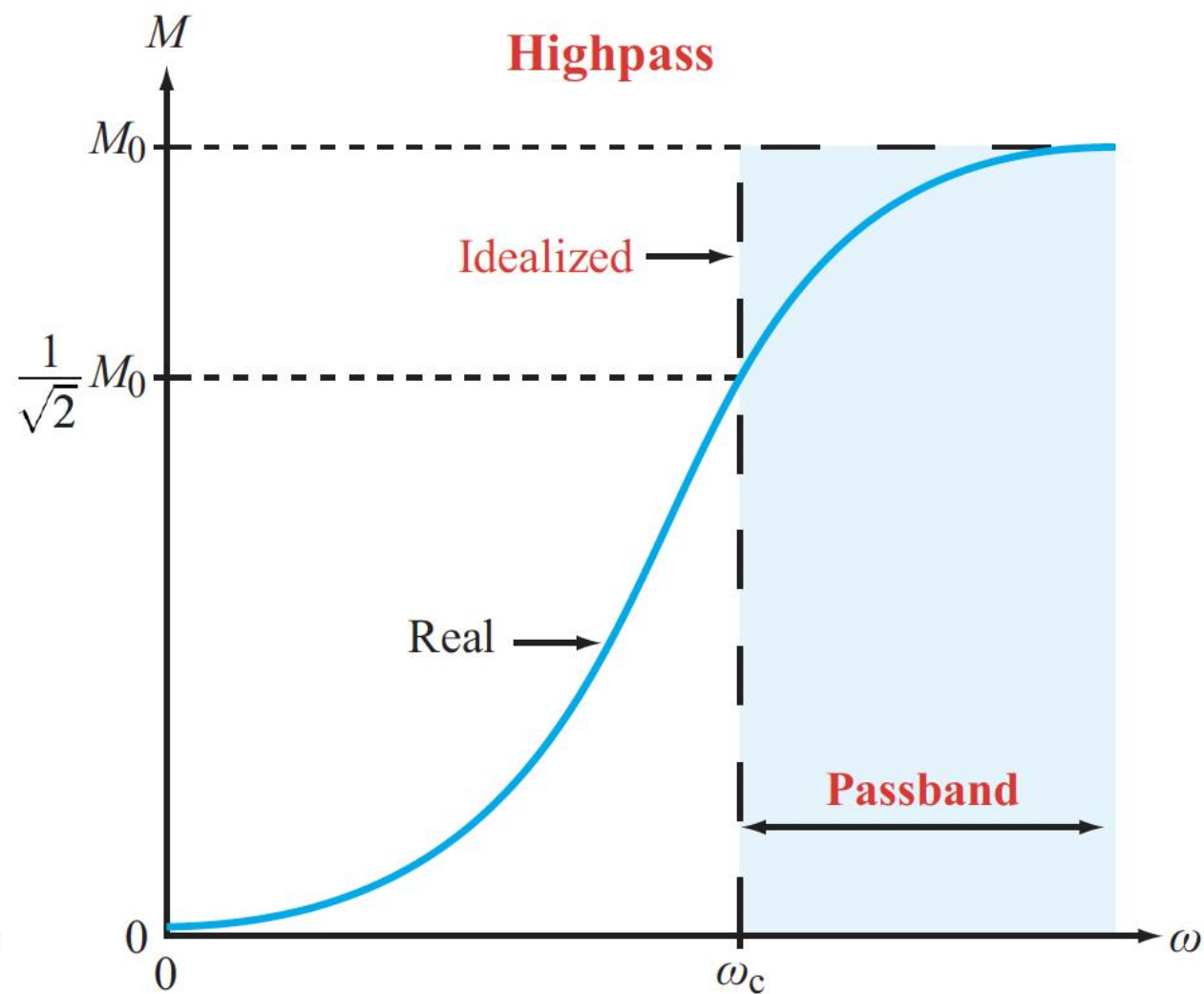
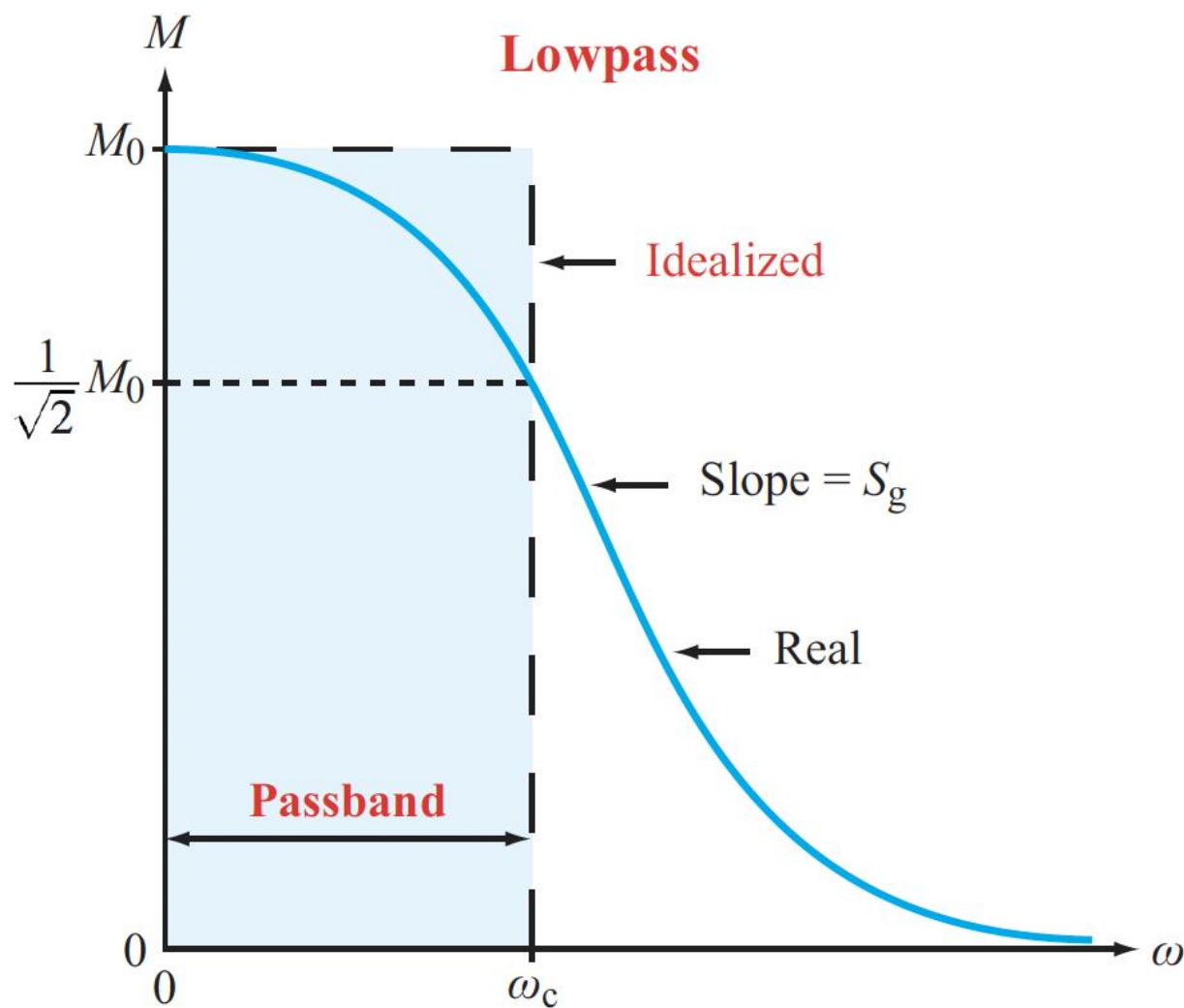
Lecture 24: Fourier Applications

**Reference: Chapter 6
(Section 6-2)**

Please turn off your gadgets now

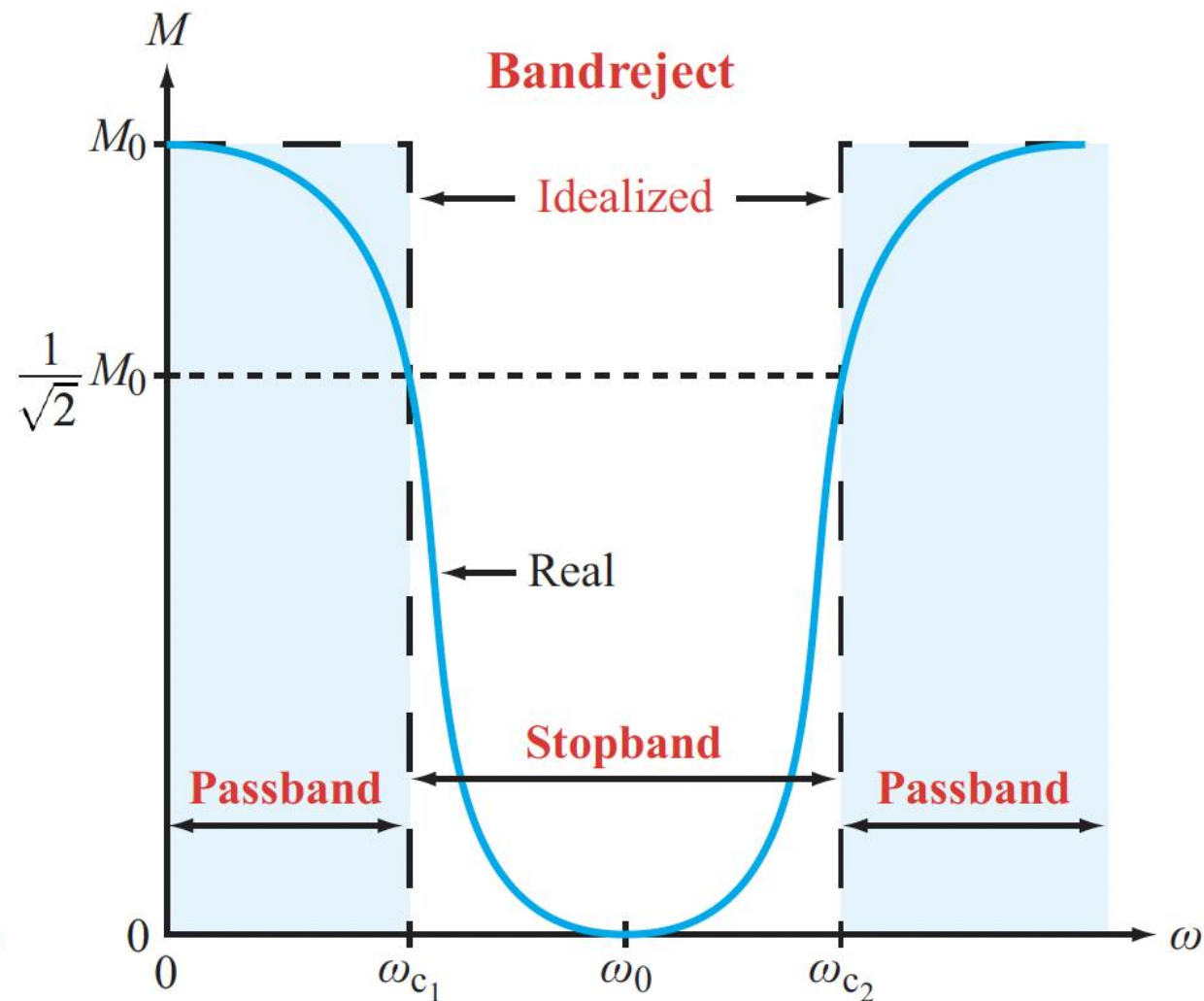
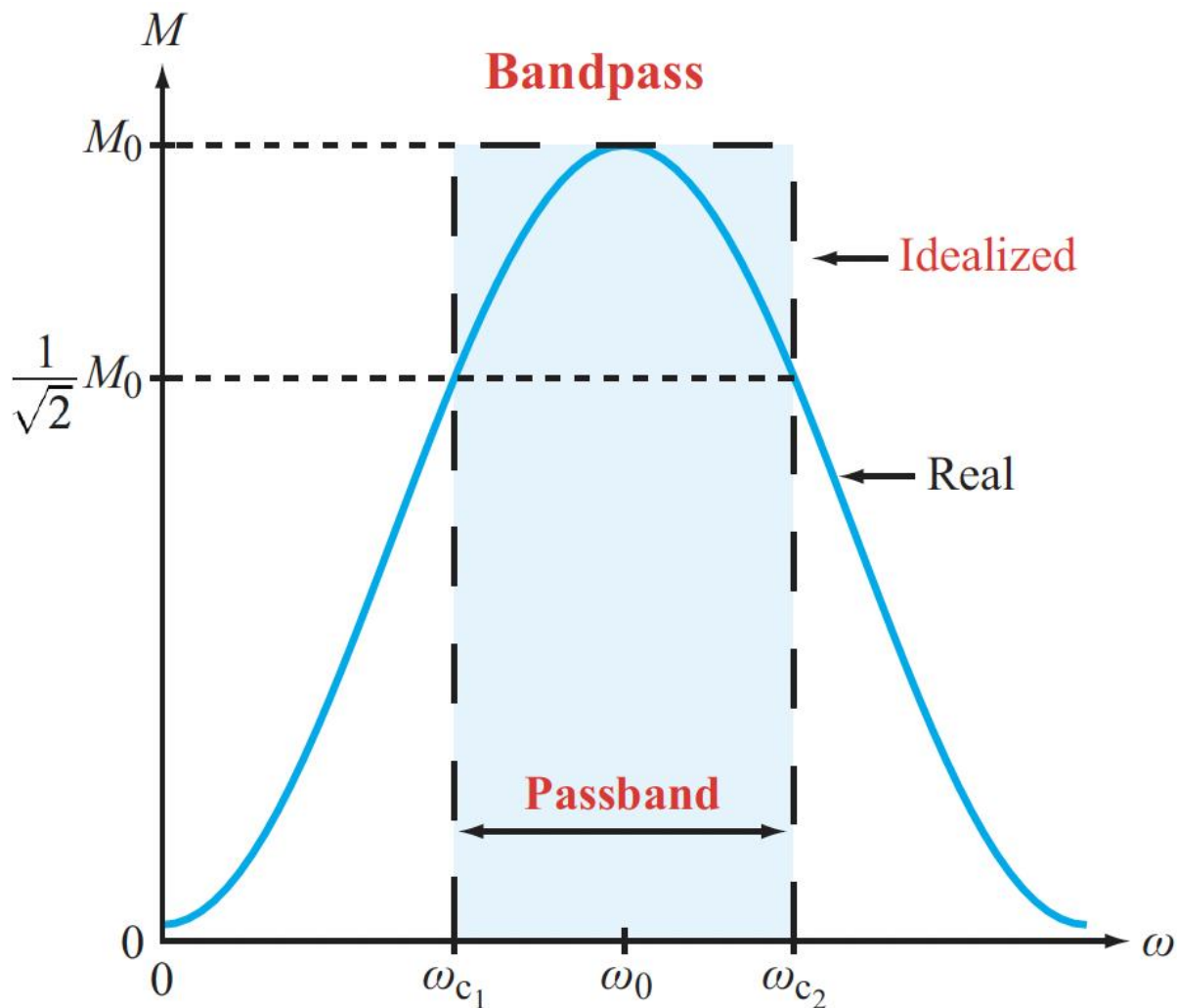


Recap: Common Basic Filters



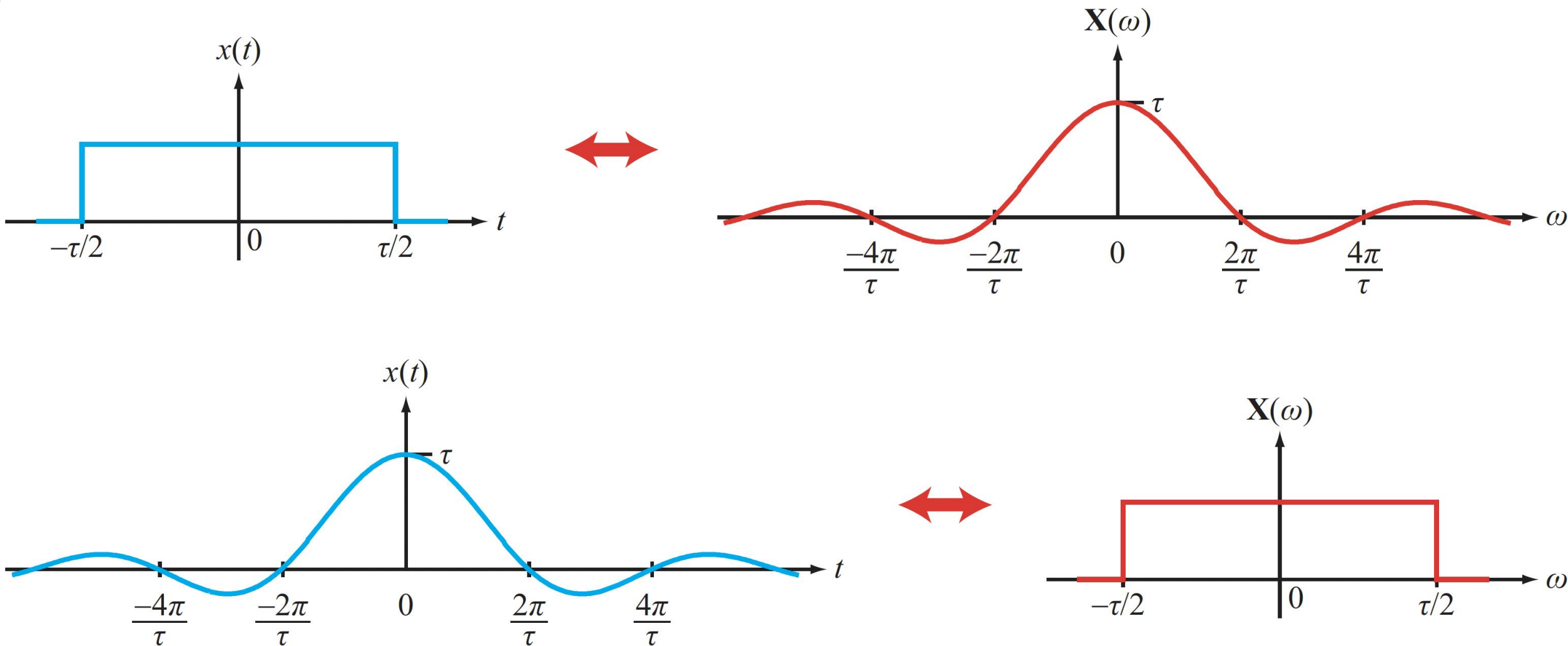


Recap: Common Basic Filters





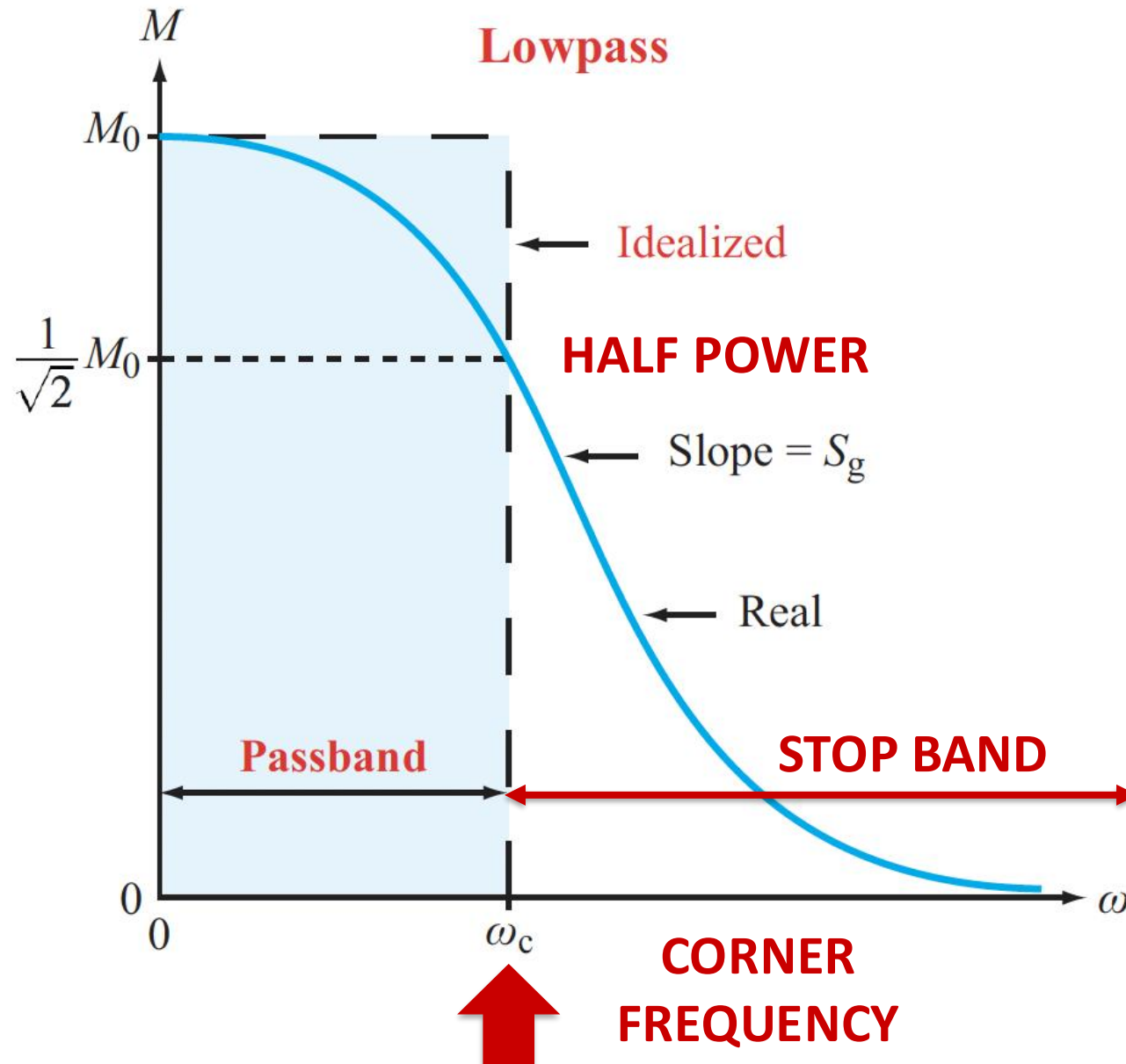
The Pulse $\leftrightarrow$ Fourier Transform



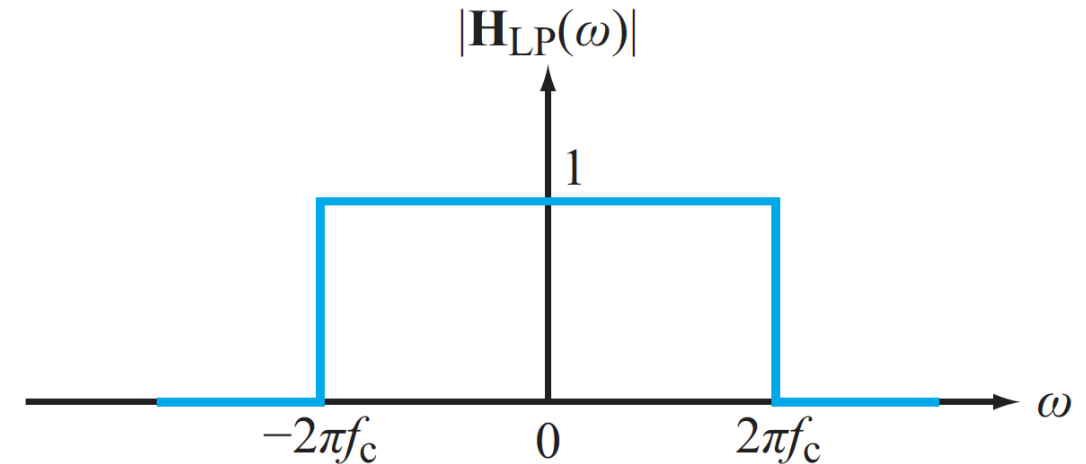
Duality in action



The Low-pass Filter



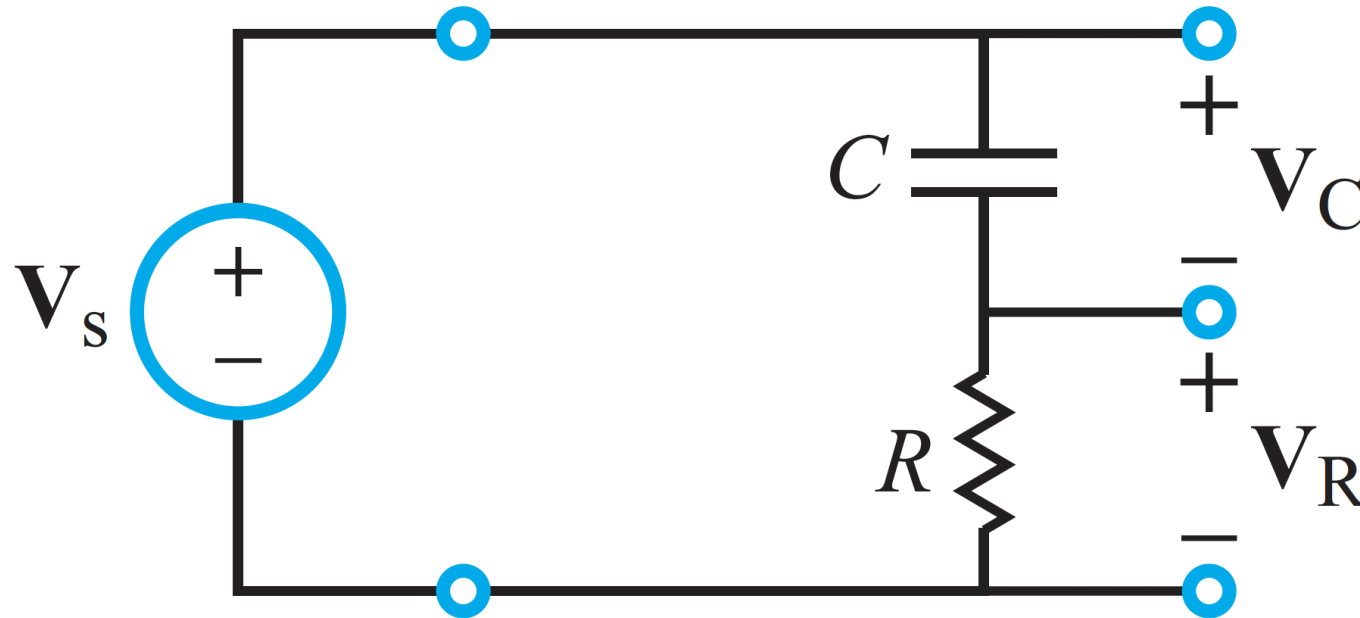
The ideal low-pass filter is a “pulse” in the frequency domain



- Unfortunately, the time-domain inverse (sinc function) is noncausal, so an ideal Low-pass filter is impossible to build
- This is also true for the four other idealized filters



The RC Circuit as a Filter



$$\mathbf{Z}_C = \frac{1}{j\omega C}$$

**SHORT CIRCUIT
AT HIGH FREQUENCY**

$$\mathbf{Z}_R = R$$

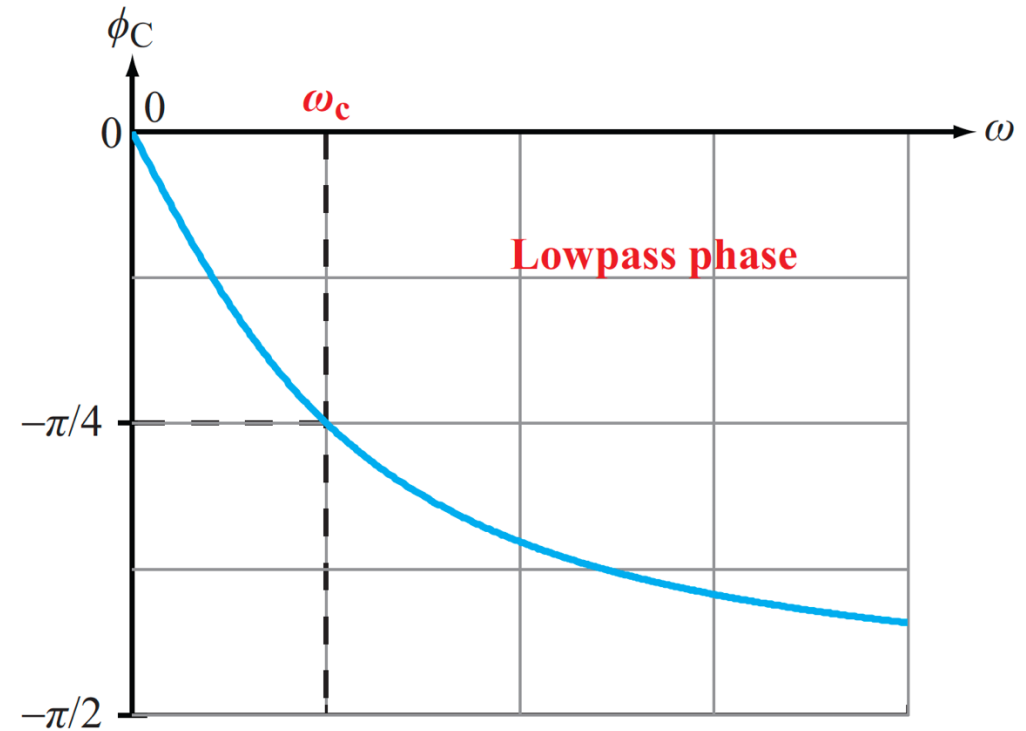
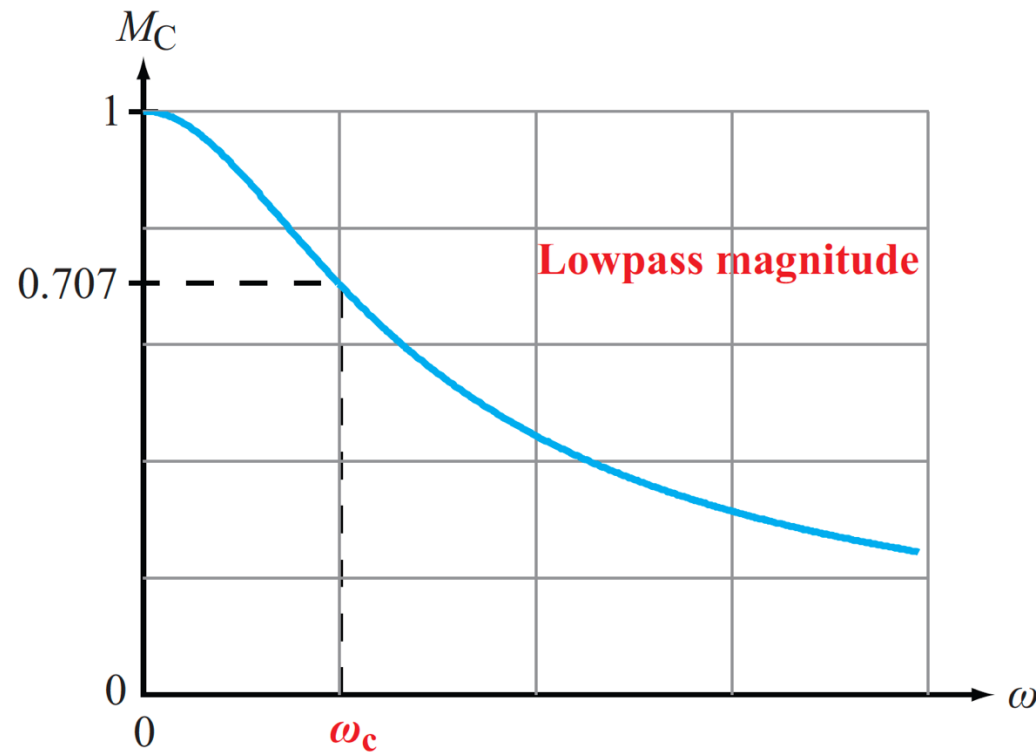
**FREQUENCY
INDEPENDENT**

$$\mathbf{V}_C = \mathbf{V}_S \left(\frac{\mathbf{Z}_C}{\mathbf{Z}_C + R} \right) \Rightarrow \hat{\mathbf{H}}_C(\omega) = \frac{\mathbf{V}_C}{\mathbf{V}_S} = \frac{1}{(1 + j\omega RC)}$$

$$\mathbf{V}_R = \mathbf{V}_S \left(\frac{R}{\mathbf{Z}_C + R} \right) \Rightarrow \hat{\mathbf{H}}_R(\omega) = \frac{\mathbf{V}_R}{\mathbf{V}_S} = \frac{j\omega RC}{(1 + j\omega RC)}$$



Low-pass Response



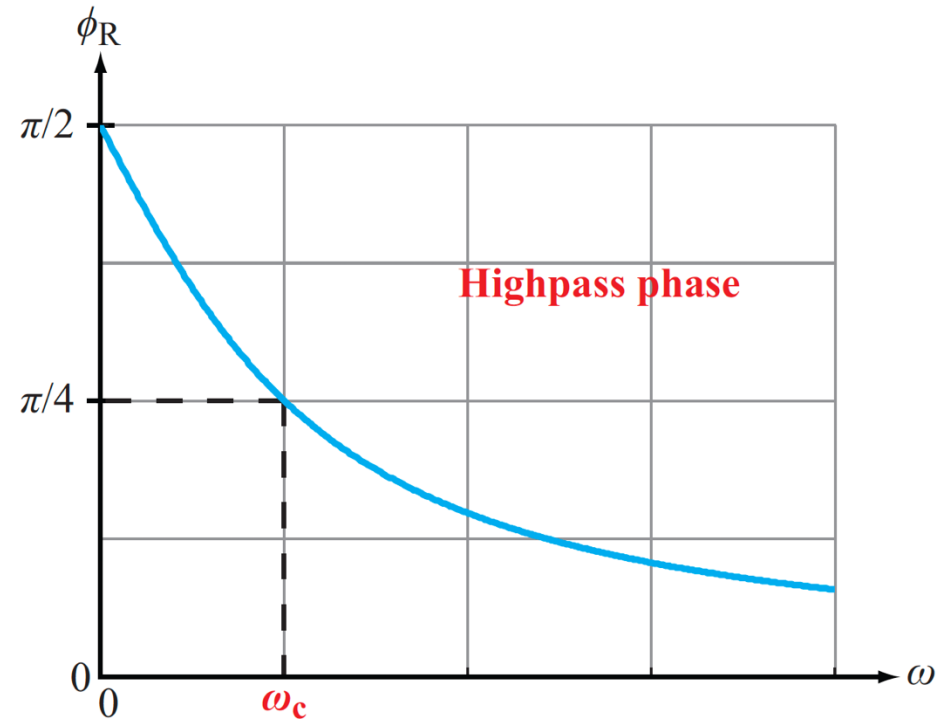
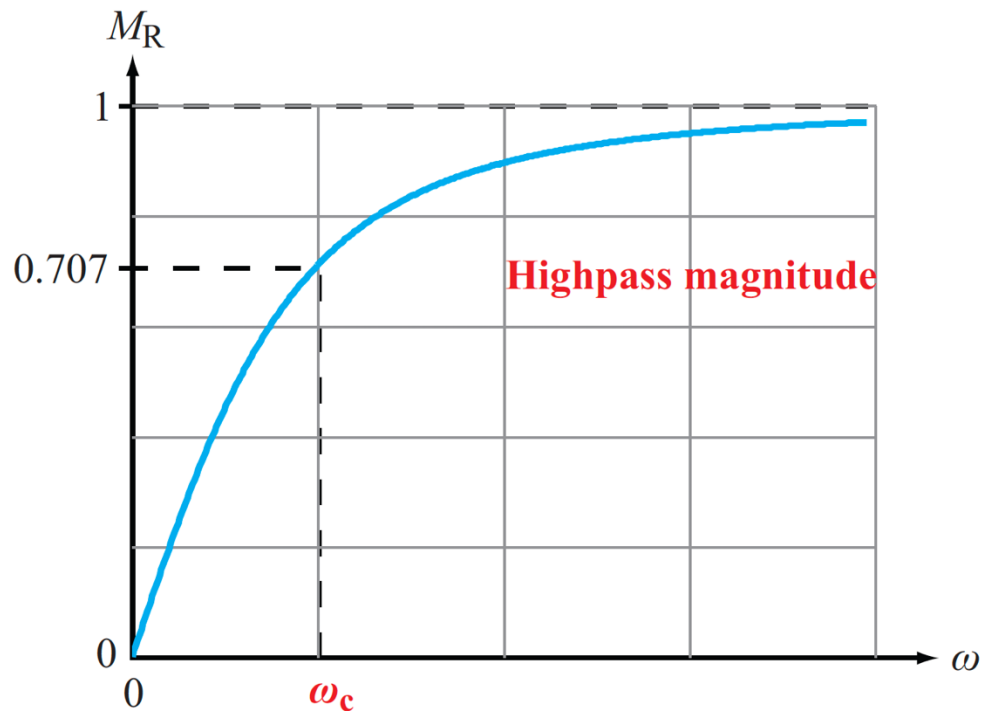
$$M_C(\omega) = |\hat{\mathbf{H}}_C(\omega)| = \left| \frac{1}{(1 + j\omega RC)} \right| = \frac{|1|}{|1 + j\omega RC|} = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}, \quad \omega_c = \frac{1}{RC}$$

$$\phi_C(\omega) = \phi(1) - \phi(1 + j\omega RC) = 0 - \tan^{-1}(\omega RC)$$

LET'S TALK ABOUT WHAT HAPPENS AS THE RC VALUE IS VARIED



High-pass Response



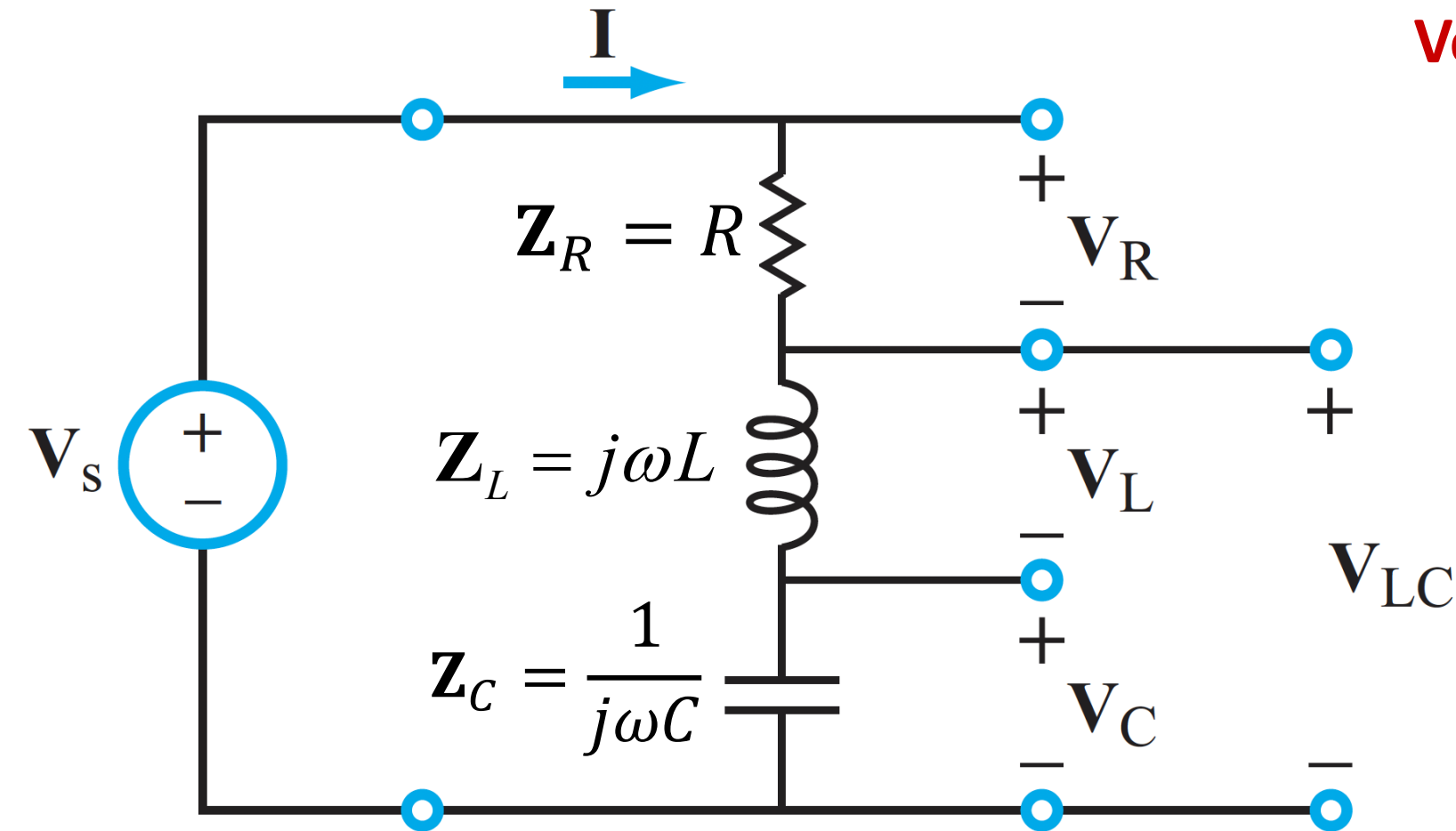
$$M_R(\omega) = |\hat{\mathbf{H}}_R(\omega)| = \left| \frac{j\omega RC}{(1 + j\omega RC)} \right| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} = \frac{\left(\frac{\omega}{\omega_c} \right)}{\sqrt{1 + \left(\frac{\omega}{\omega_c} \right)^2}}, \quad \omega_c = \frac{1}{RC}$$

$$\phi_R(\omega) = \phi(j\omega RC) - \phi(1 + j\omega RC) = \frac{\pi}{2} - \tan^{-1}(\omega RC)$$

**LET'S TALK ABOUT
WHAT HAPPENS AS THE
RC VALUE IS VARIED**



RLC Circuit Example



Voltage divider structure

$$\hat{H}_R(\omega) = \frac{Z_R}{Z_R + Z_C + Z_L}$$

$$\hat{H}_L(\omega) = \frac{Z_L}{Z_R + Z_C + Z_L}$$

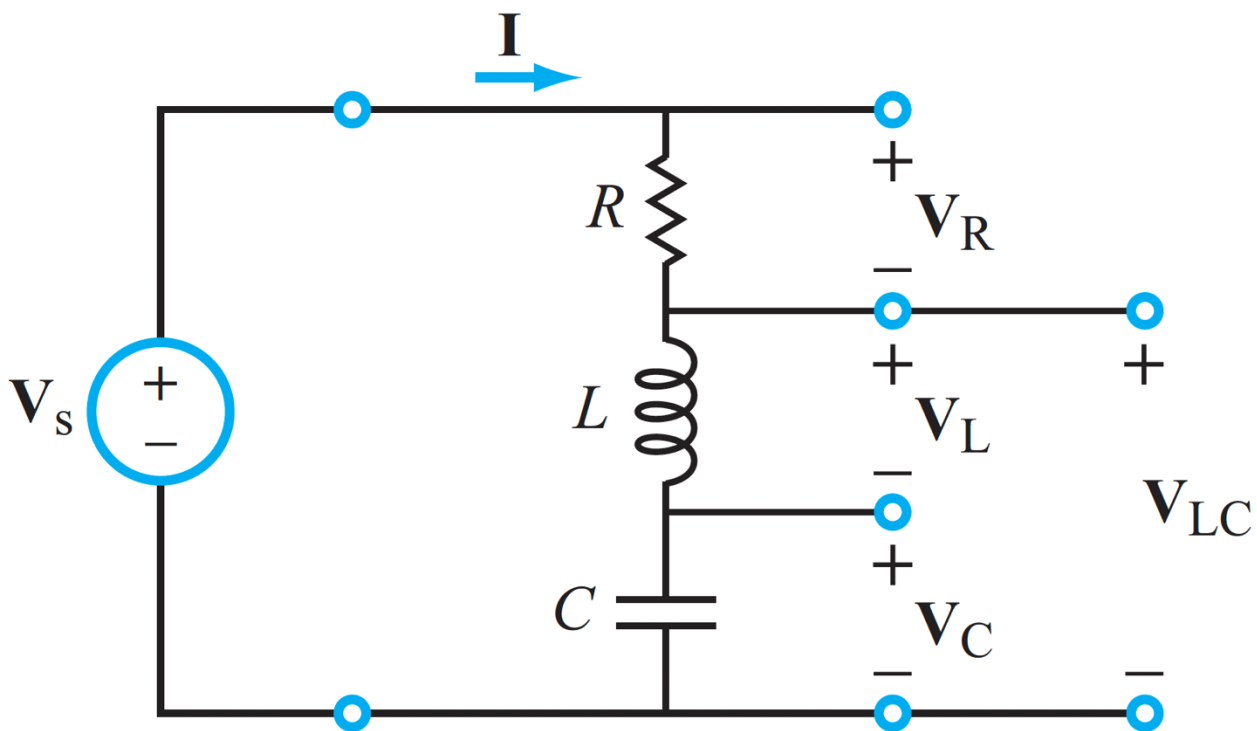
$$\hat{H}_C(\omega) = \frac{Z_C}{Z_R + Z_C + Z_L}$$

$$\hat{H}_{LC}(\omega) = \frac{Z_C + Z_L}{Z_R + Z_C + Z_L}$$

Each of the possible outputs defines a different filter



RLC Circuit Example



$$\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L + \frac{1}{j\omega C}} = \frac{j\omega C \mathbf{V}_s}{j\omega RC + (1 - \omega^2 LC)}$$

$$\hat{\mathbf{H}}_R(\omega) = \frac{\mathbf{V}_R}{\mathbf{V}_s} = \frac{\mathbf{I}R}{\mathbf{V}_s} = \frac{j\omega RC}{j\omega RC + (1 - \omega^2 LC)}$$

Interesting Observation:

$$\mathbf{Z}_L + \mathbf{Z}_C = j\omega L + \frac{1}{j\omega C} = j\left(\omega L - \frac{1}{\omega C}\right)$$

This can become 0 when $\omega L = \frac{1}{\omega C}$ or $\omega = \frac{1}{\sqrt{LC}}$

From this, can we guess what kind of filter $\hat{\mathbf{H}}_{LC}(\omega)$ is?



Resistor Voltage V_R

$$|\hat{\mathbf{H}}_R(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

when $\omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \rightarrow 0$

when $\omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \rightarrow 0$

The denominator is minimum,

or $|\hat{\mathbf{H}}_R(\omega)|$ is maximum

when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

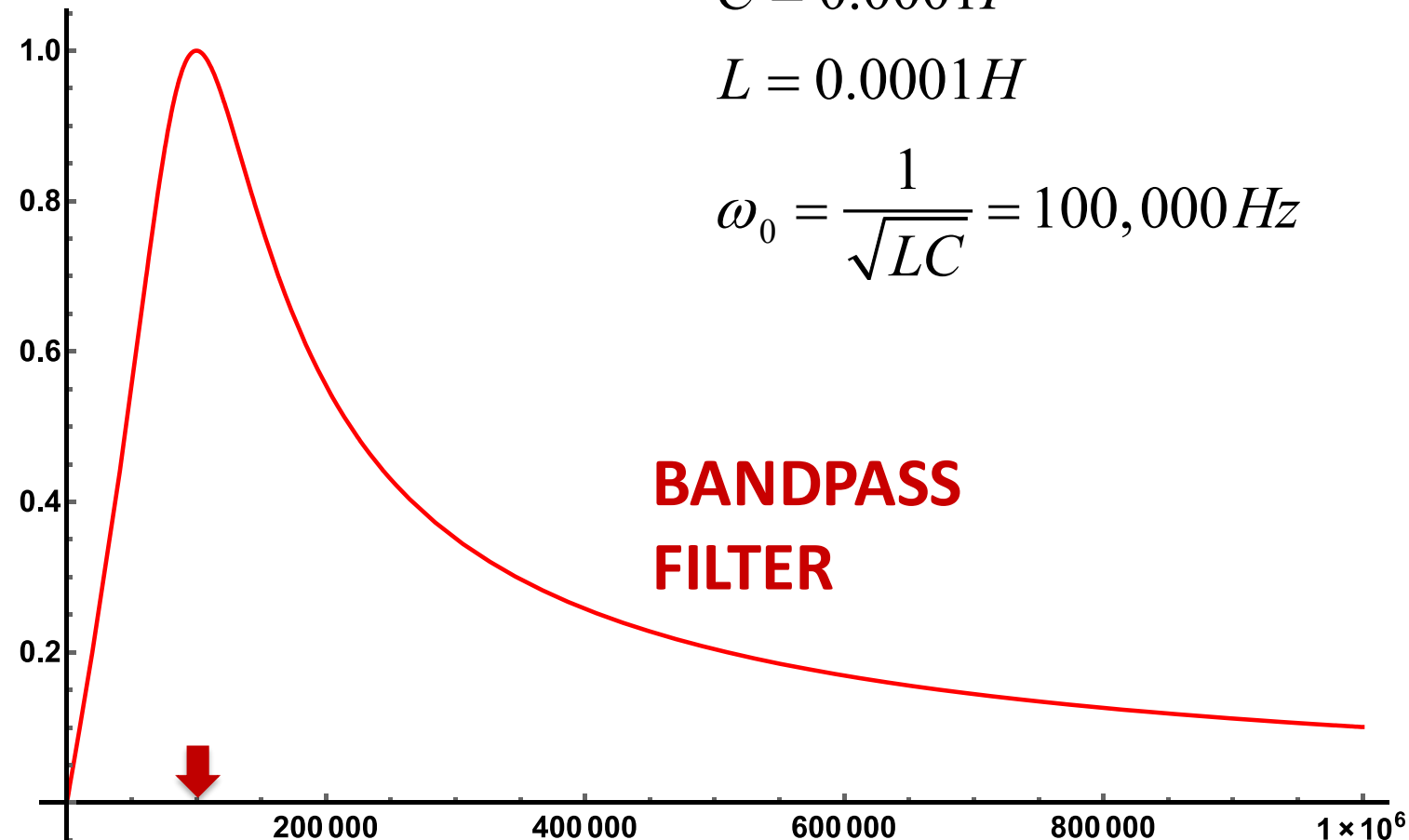
Example:

$$R = 1\Omega$$

$$C = 0.0001F$$

$$L = 0.0001H$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 100,000Hz$$





Resistor Voltage V_R

$$|\hat{\mathbf{H}}_R(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

when $\omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \rightarrow 0$

when $\omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \rightarrow 0$

The denominator is minimum,

or $|\hat{\mathbf{H}}_R(\omega)|$ is maximum

when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

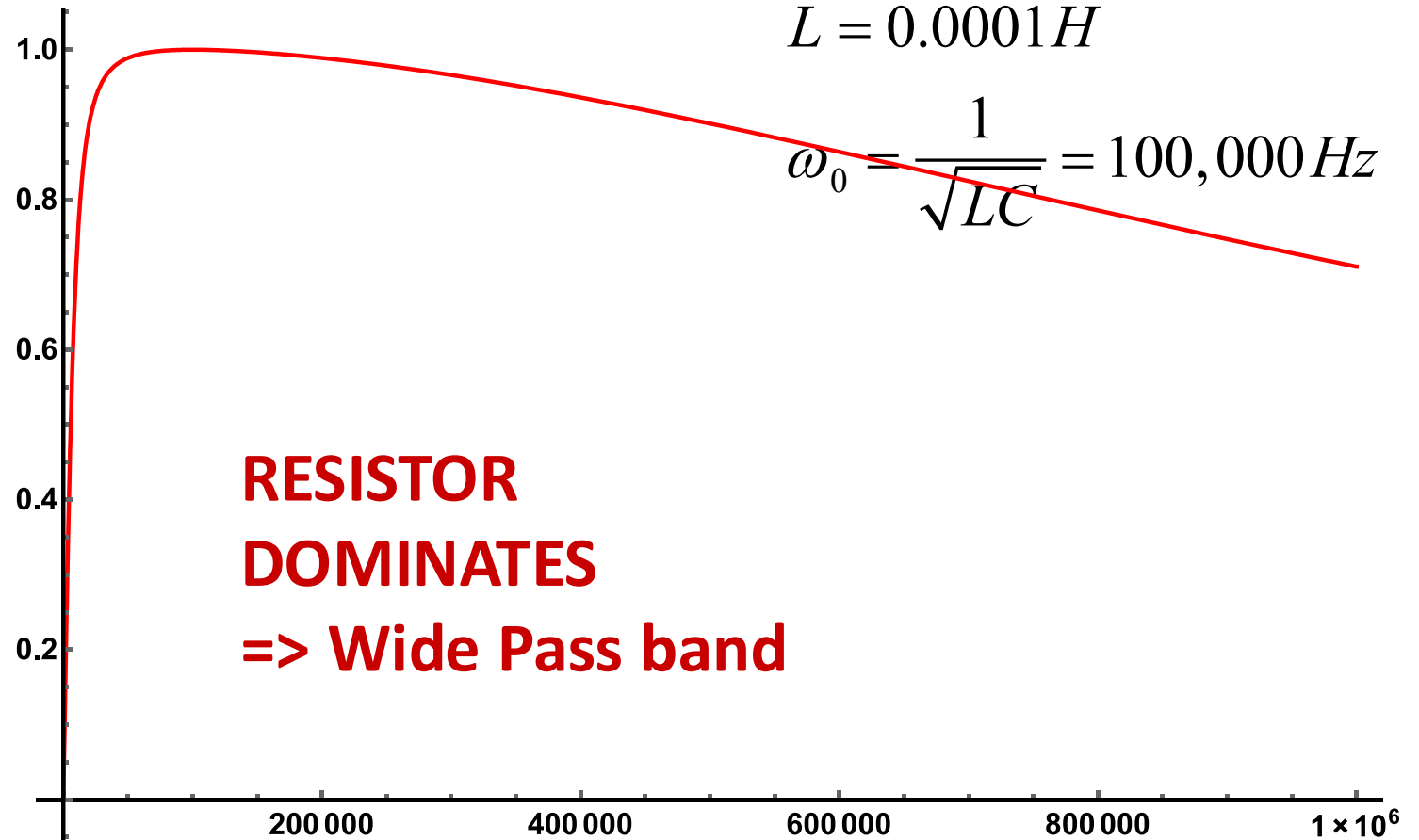
Example:

$R = 10\Omega$ 

$C = 0.0001F$

$L = 0.0001H$

$\omega_0 = \frac{1}{\sqrt{LC}} = 100,000Hz$





Resistor Voltage V_R

$$|\hat{\mathbf{H}}_R(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

$$\text{when } \omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \rightarrow 0$$

$$\text{when } \omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \rightarrow 0$$

The denominator is minimum,

or $|\hat{\mathbf{H}}_R(\omega)|$ is maximum

when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

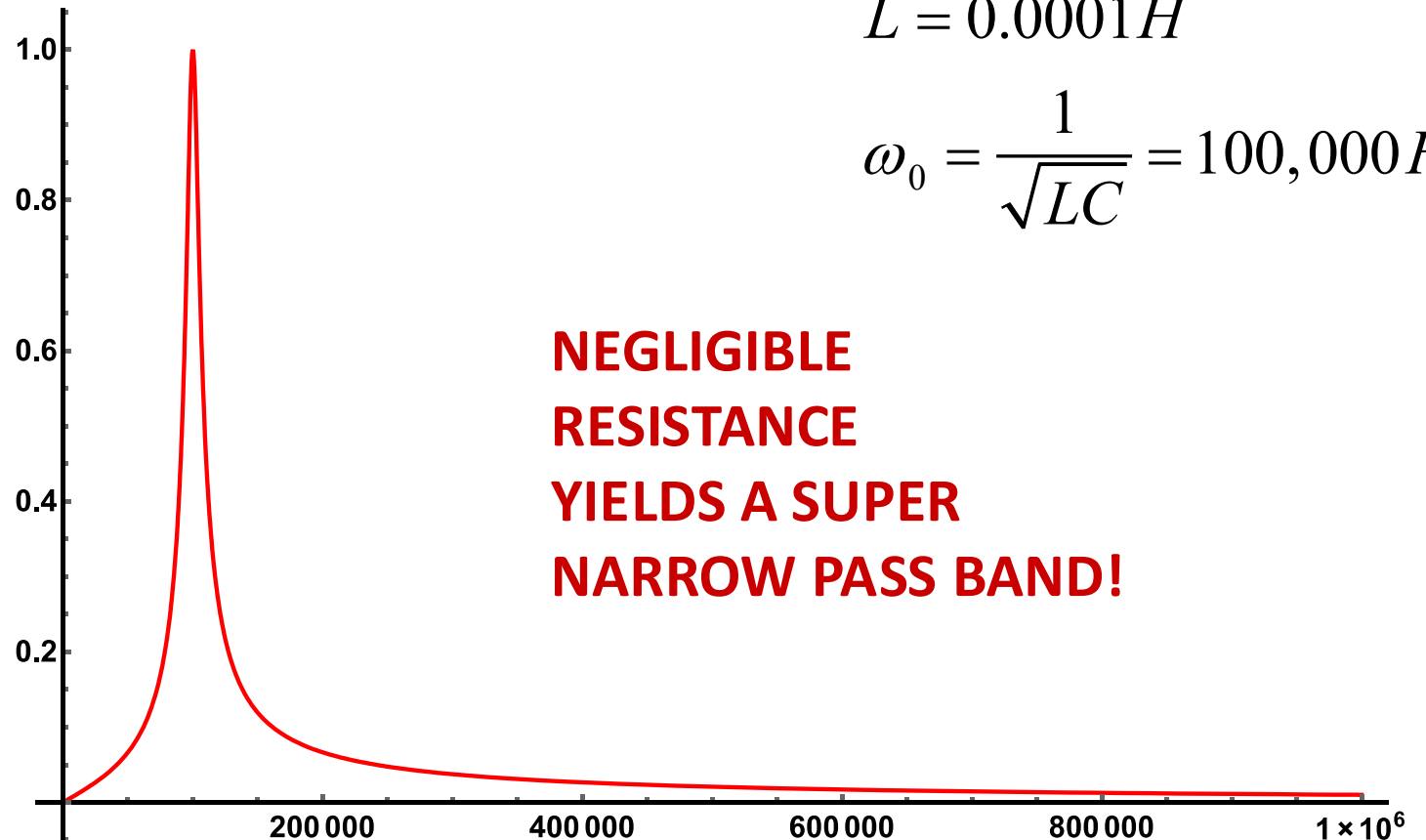
Example:

$$R = 0.1\Omega$$

$$C = 0.0001F$$

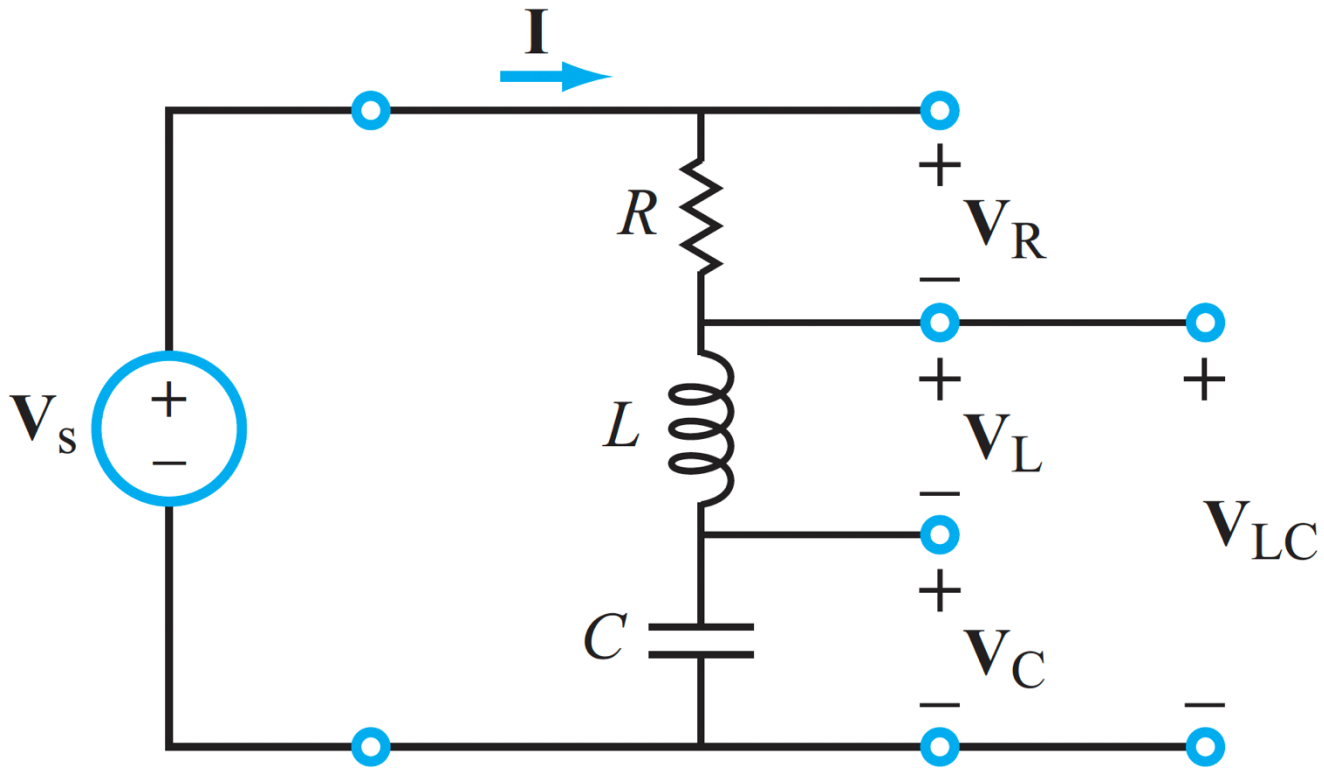
$$L = 0.0001H$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 100,000 Hz$$





Inductor Voltage V_L



$$\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L + \frac{1}{j\omega C}} = \frac{j\omega C \mathbf{V}_s}{j\omega RC + (1 - \omega^2 LC)}$$

$$\hat{\mathbf{H}}_L(\omega) = \frac{\mathbf{V}_L}{\mathbf{V}_s} = \frac{(j\omega L)\mathbf{I}}{\mathbf{V}_s} = \frac{(j\omega L)(j\omega C)}{j\omega RC + (1 - \omega^2 LC)}$$

$$|\hat{\mathbf{H}}_L(\omega)| = \frac{\omega^2 LC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$



Inductor Voltage V_L

$$|\hat{\mathbf{H}}_L(\omega)| = \frac{\omega^2 LC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

when $\omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \rightarrow 0$

when $\omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \rightarrow 1$

The denominator is minimum,

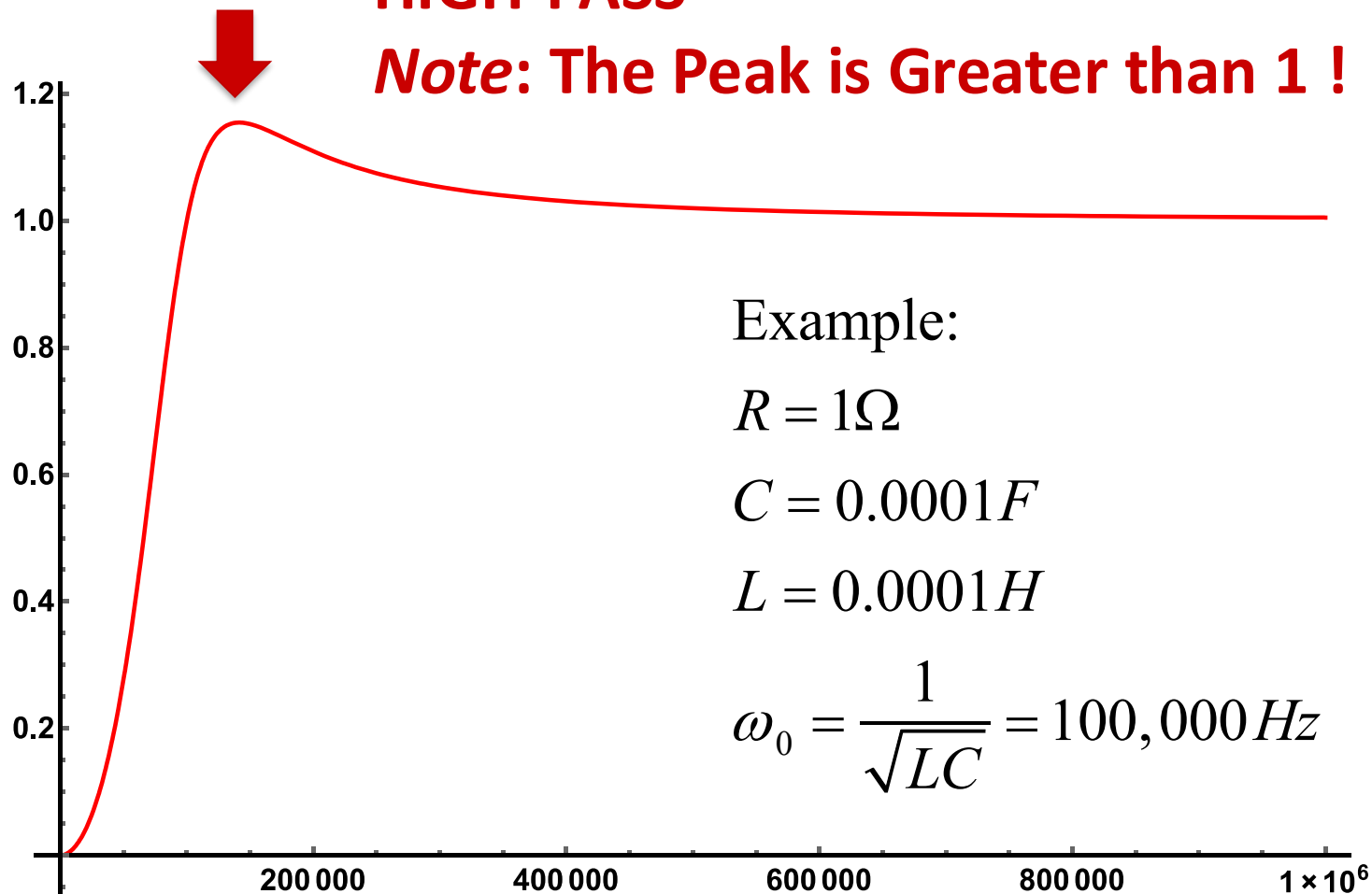
or $|\hat{\mathbf{H}}_L(\omega)|$ is maximum

when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \rightarrow \frac{1}{R} \sqrt{\frac{L}{C}}$$

HIGH-PASS

Note: The Peak is Greater than 1 !



Example:

$$R = 1\Omega$$

$$C = 0.0001F$$

$$L = 0.0001H$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 100,000Hz$$



Inductor Voltage V_L

$$|\hat{\mathbf{H}}_L(\omega)| = \frac{\omega^2 LC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

$$\text{when } \omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \rightarrow 0$$

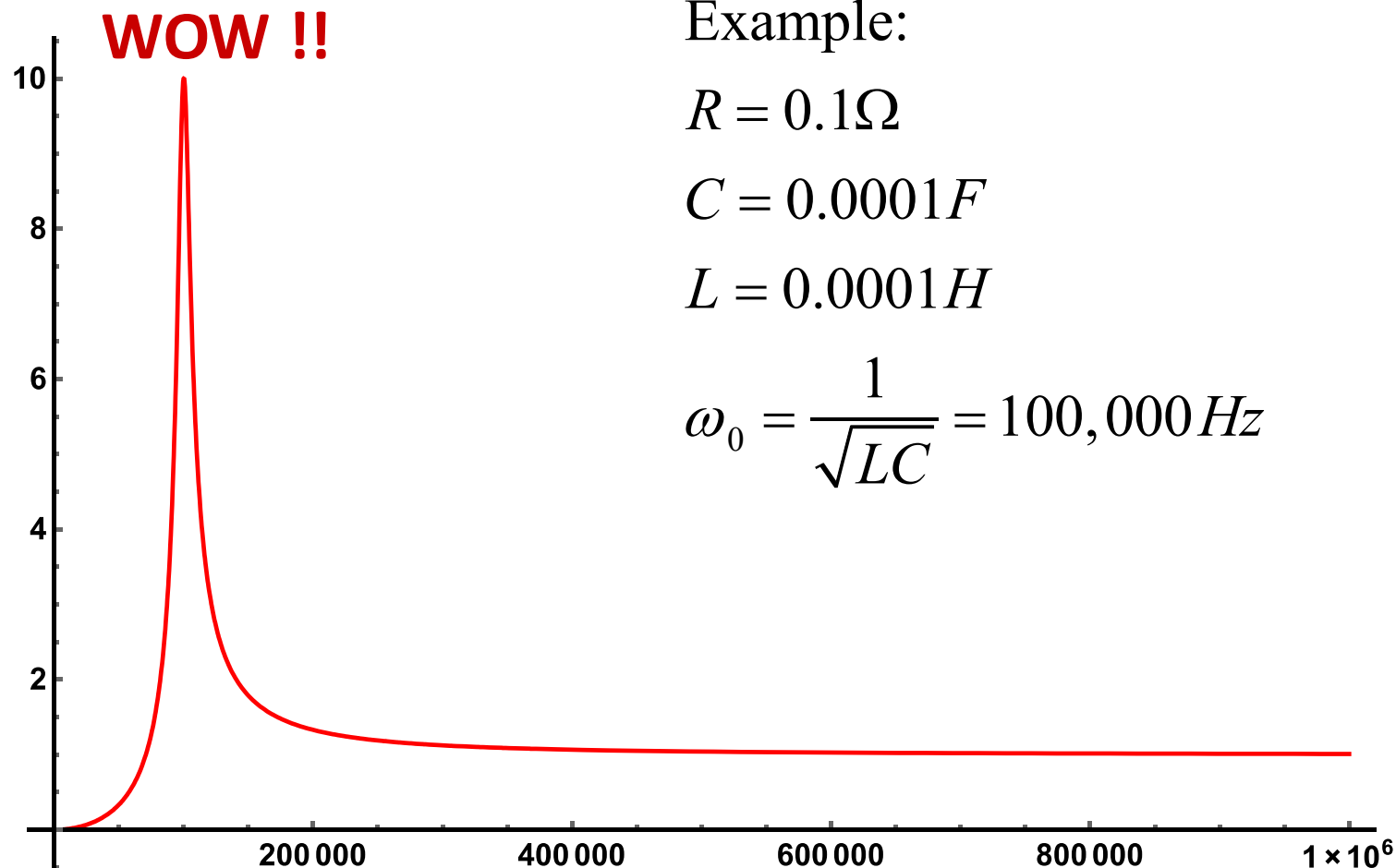
$$\text{when } \omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \rightarrow 1$$

The denominator is minimum,

or $|\hat{\mathbf{H}}_L(\omega)|$ is maximum

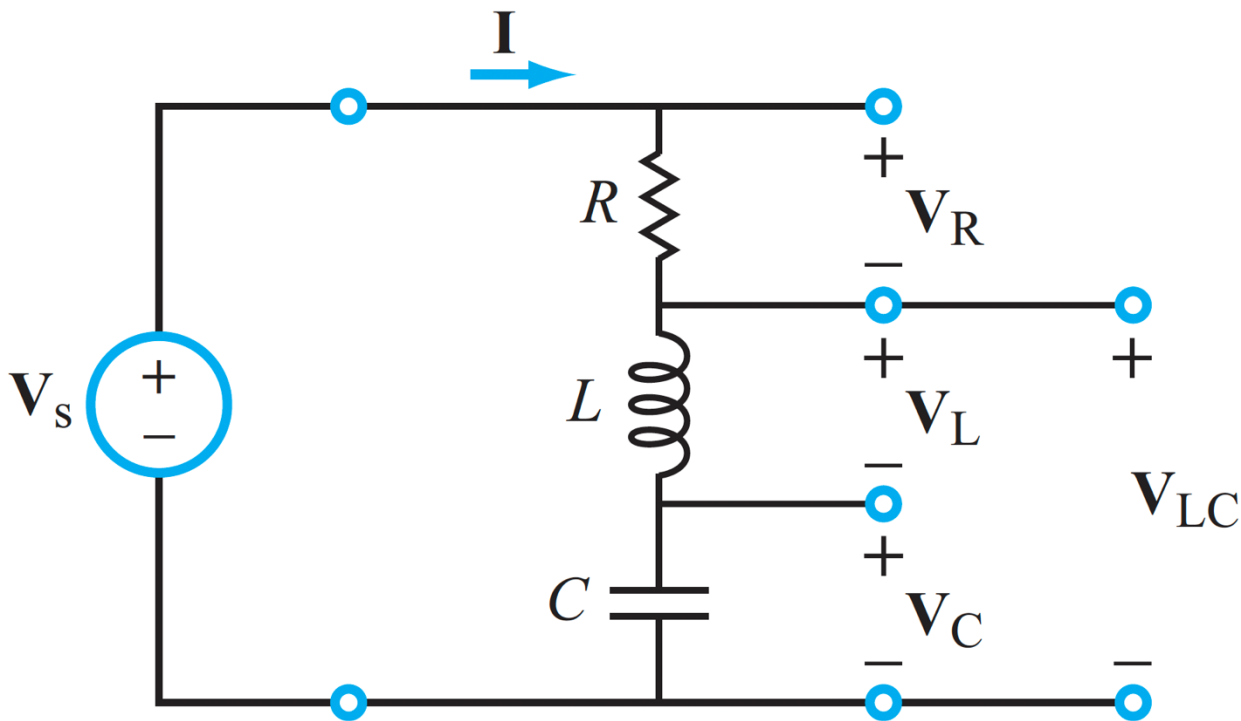
when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \rightarrow \frac{1}{R} \sqrt{\frac{L}{C}}$$





Capacitor Voltage V_C



$$\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L + \frac{1}{j\omega C}} = \frac{j\omega C \mathbf{V}_s}{j\omega RC + (1 - \omega^2 LC)}$$

$$\hat{\mathbf{H}}_C(\omega) = \frac{\mathbf{V}_C}{\mathbf{V}_s} = \frac{\mathbf{I}}{j\omega C} \square \frac{1}{\mathbf{V}_s} = \frac{1}{j\omega RC + (1 - \omega^2 LC)}$$

$$|\hat{\mathbf{H}}_C(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$



Capacitor Voltage V_C

$$|\hat{\mathbf{H}}_C(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

$$\text{when } \omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_C(\omega)| \rightarrow 1$$

$$\text{when } \omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_C(\omega)| \rightarrow 0$$

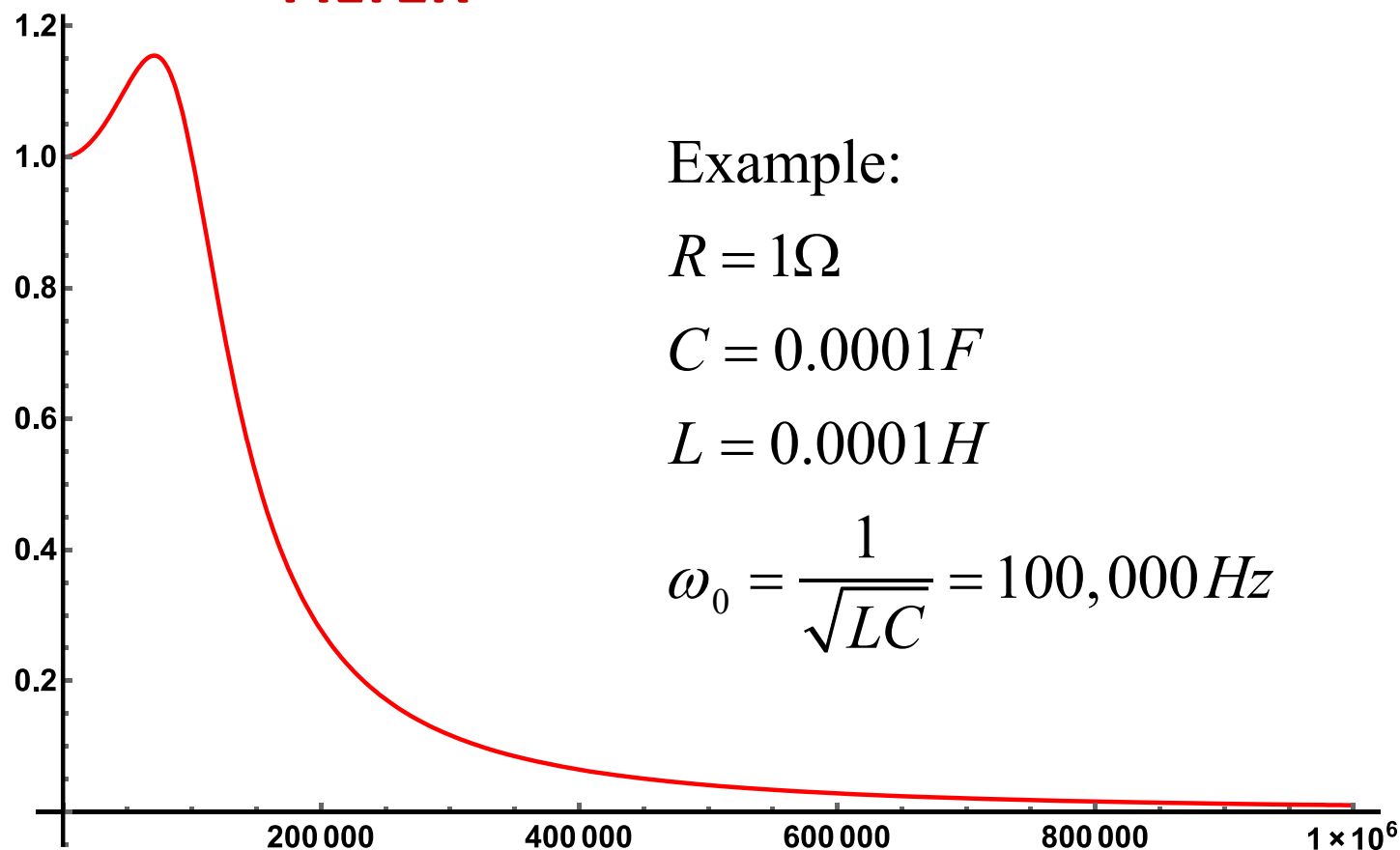
The denominator is minimum,

or $|\hat{\mathbf{H}}_C(\omega)|$ is maximum

$$\text{when } \omega^2 LC = 1,$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow |\mathbf{H}_C(\omega)| \rightarrow \frac{1}{R} \sqrt{\frac{L}{C}}$$

LOW-PASS FILTER





Capacitor Voltage V_C

$$|\hat{\mathbf{H}}_C(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

$$\text{when } \omega \rightarrow 0 \Rightarrow |\hat{\mathbf{H}}_C(\omega)| \rightarrow 1$$

$$\text{when } \omega \rightarrow \infty \Rightarrow |\hat{\mathbf{H}}_C(\omega)| \rightarrow 0$$

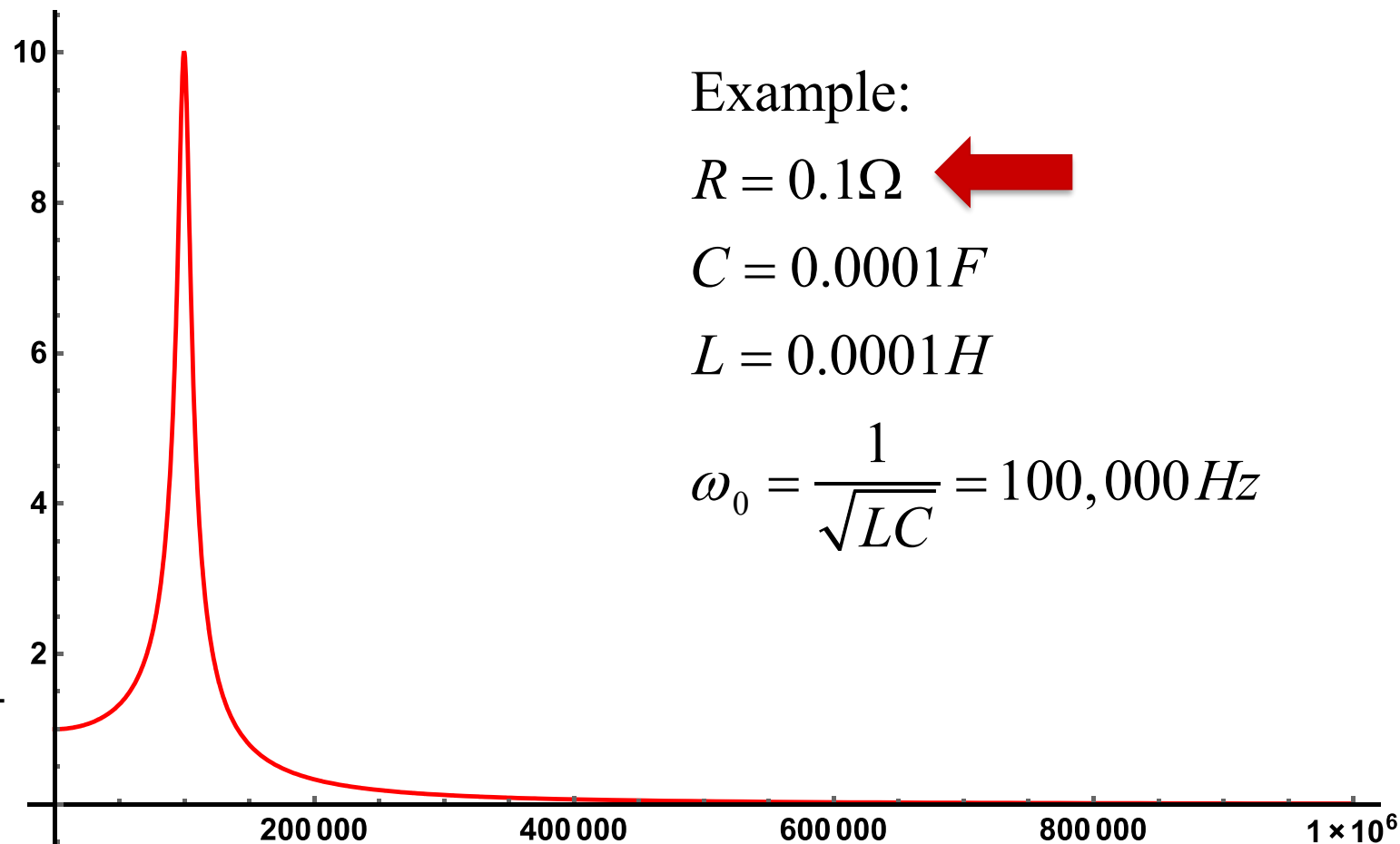
The denominator is minimum,

or $|\hat{\mathbf{H}}_C(\omega)|$ is maximum

when $\omega^2 LC = 1$,

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow |\mathbf{H}_C(\omega)| \rightarrow \frac{1}{R} \sqrt{\frac{L}{C}}$$

That's Wild !!



Example:

$$R = 0.1 \Omega$$



$$C = 0.0001 F$$

$$L = 0.0001 H$$

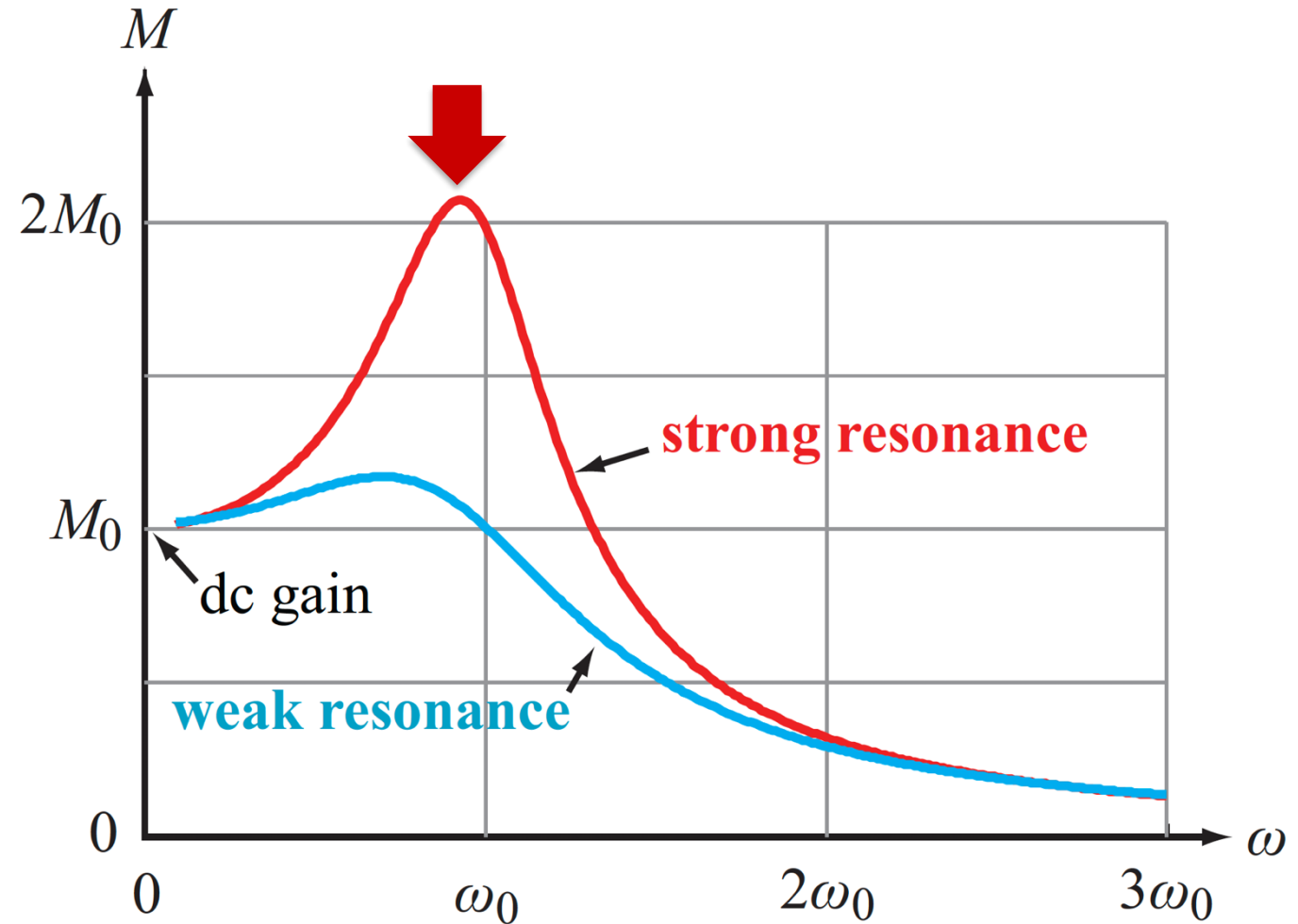
$$\omega_0 = \frac{1}{\sqrt{LC}} = 100,000 \text{ Hz}$$



Resonance

Our RLC example shows that filter responses may not be smooth

- Resonance is a non-smooth condition that arises in some filters at very specific “resonant frequencies”
- The amplitude at resonance can get much higher than M_0
 - This can be good or bad, depending upon the application

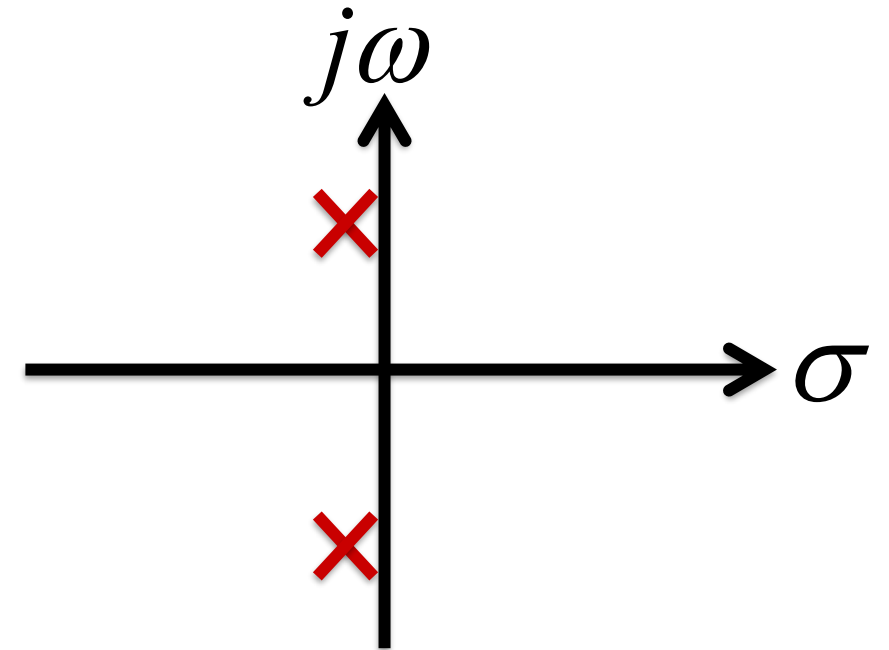




Practical Intuition

Resonance happens when 3 conditions are satisfied:

1. The circuit / system has an oscillatory mode (a pair of complex conjugate poles) with a natural frequency ω_0
2. The applied input (forcing) is at the natural frequency ω_0
3. The resistance (damping) is negligible, i.e., the system is “barely stable” and the poles are very close to, or on the vertical axis





Practical Intuition

So, how can the output exceed the input?

- Basically, resonance is not instantaneous, it is a phenomenon that accumulates over time
- At resonance, $Z_L + Z_C = 0$, so the input and output are perfectly in sync (i.e., phase shift = 0)

Bottom Line:

- Resonance is a hugely important phenomenon that occurs widely in nature
 - Can be Useful: frequency selective systems (e.g., radio)
 - Can be Dangerous: inputs at the resonance frequencies can blow up the system !



Relationship to System Modes

Consider the oscillatory system:

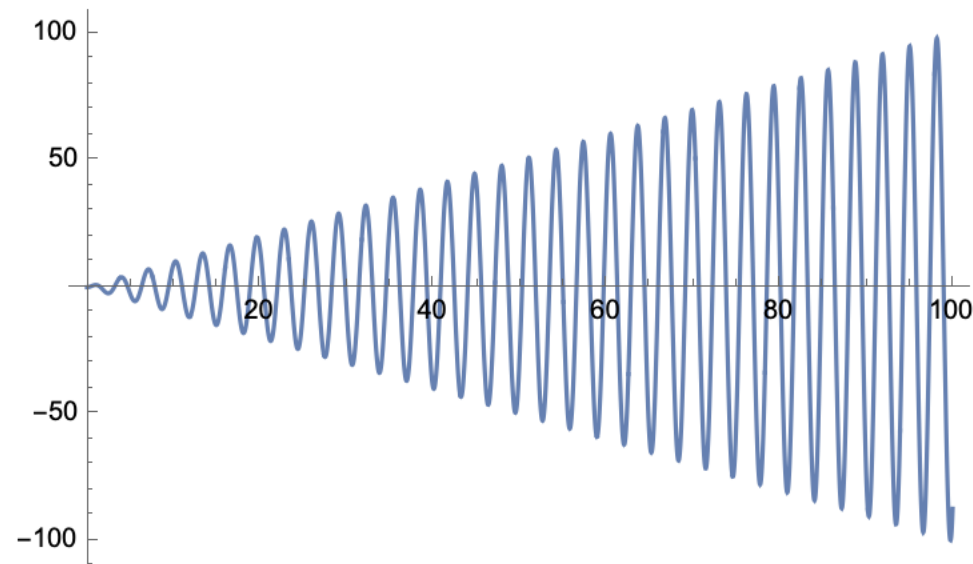
$$h(t) = e^{j\omega_2 t} u(t)$$

Let the input be $x(t) = e^{j\omega_1 t} u(t)$

$$\Rightarrow y(t) = \left[\frac{e^{j\omega_1 t} - e^{j\omega_2 t}}{j(\omega_1 - \omega_2)} \right] u(t), a \neq b$$

Clearly, the closer ω_1 is to ω_2 , the stronger the response $y(t)$ will be, and *vice versa*. In the extreme case when $\omega_1 = \omega_2 = \omega$, we get $y(t) = te^{j\omega t} u(t)$, an unbounded response!

This is “resonance” in action





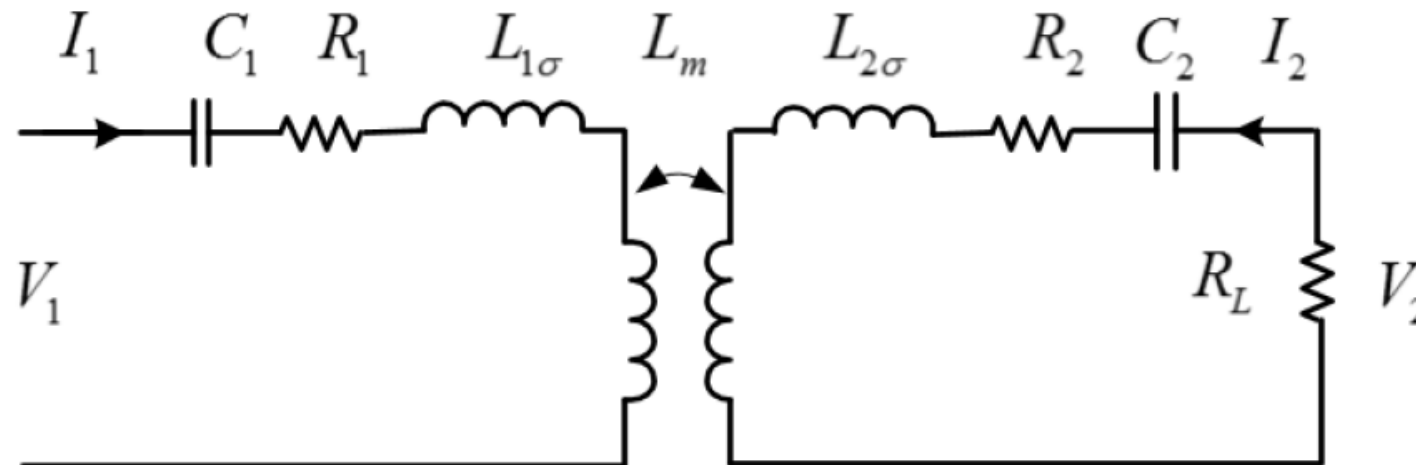
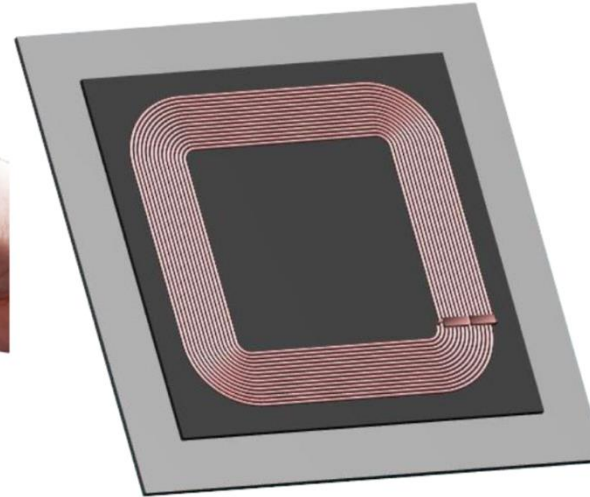
Classic Example

A sustained tone at the natural frequency of a wine glass (typically $\sim 550\text{Hz}$) can shatter it



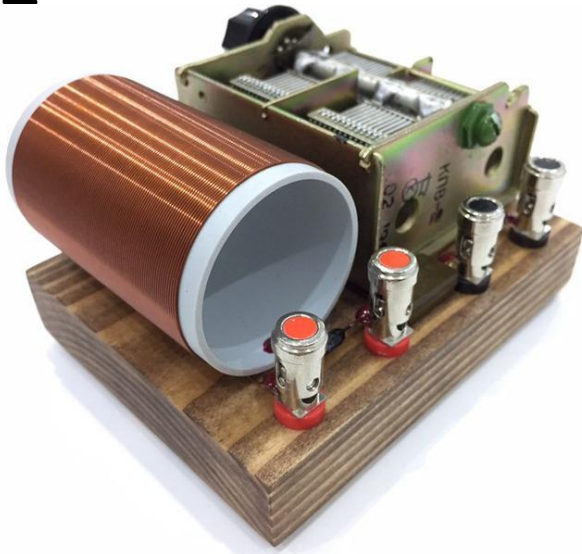
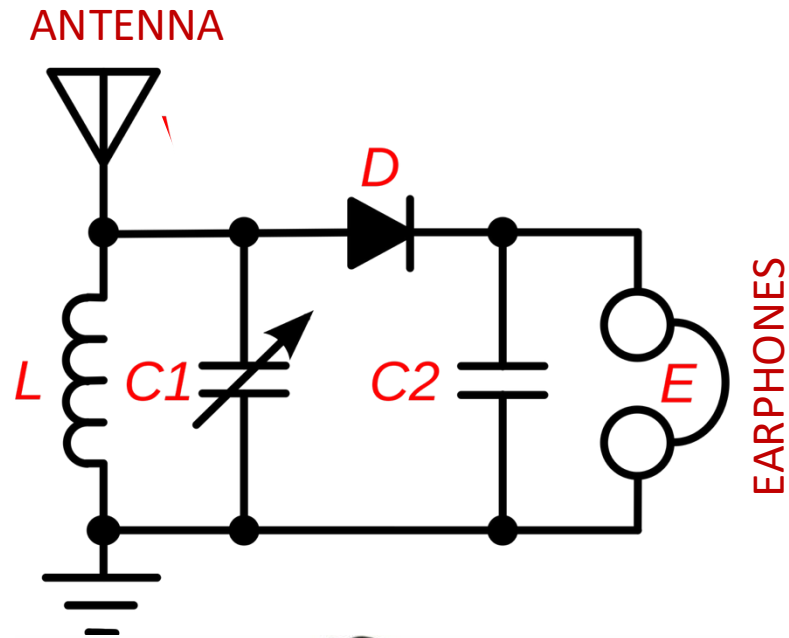


Resonant Wireless Car Charging Example

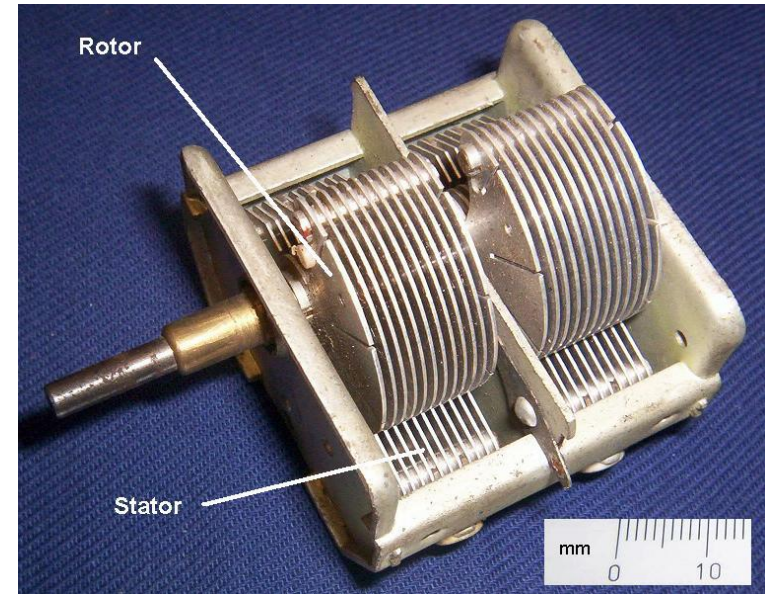




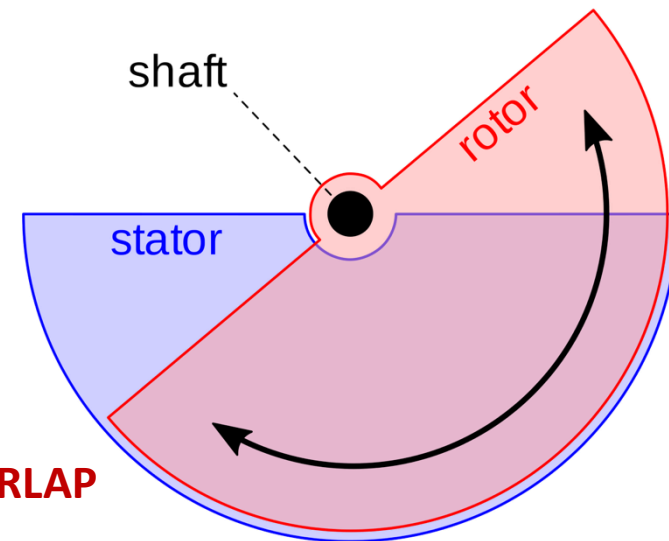
The Simplest Radio Receiver



TUNING
KNOB



VARIABLE CAPACITOR

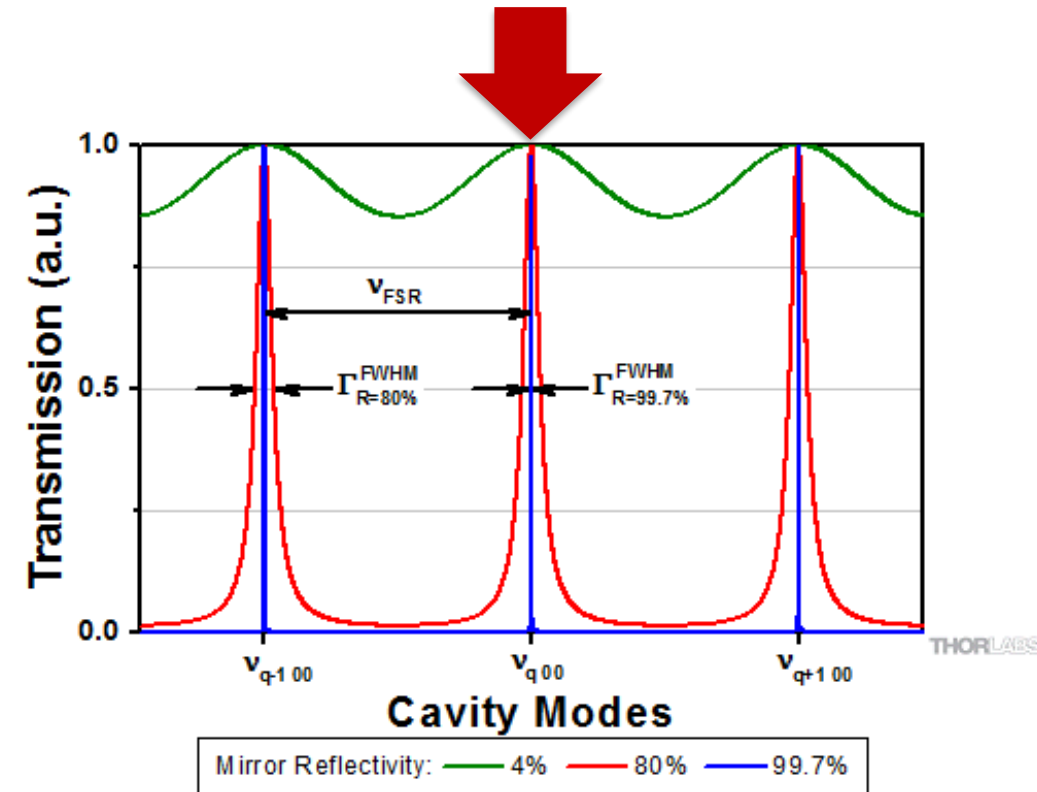
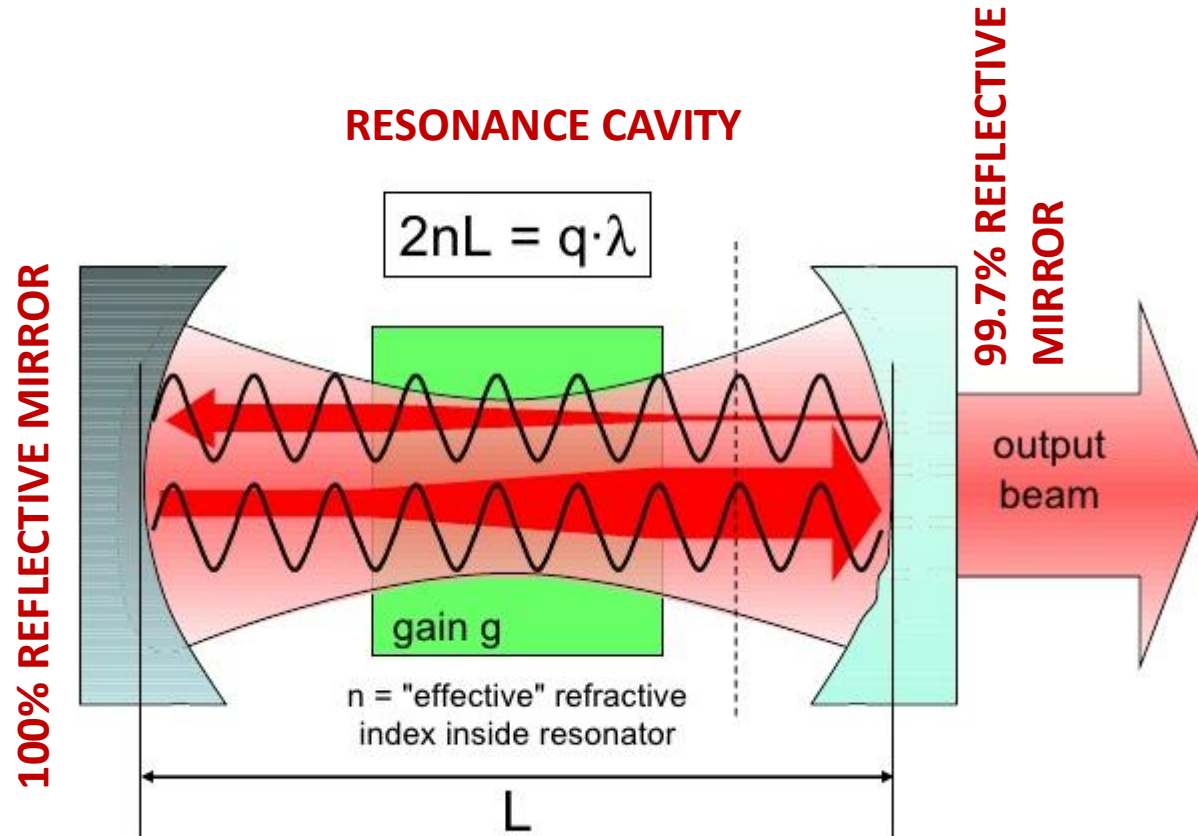


CAPACITANCE IS
PROPORTIONAL TO OVERLAP



Example: How Lasers Work

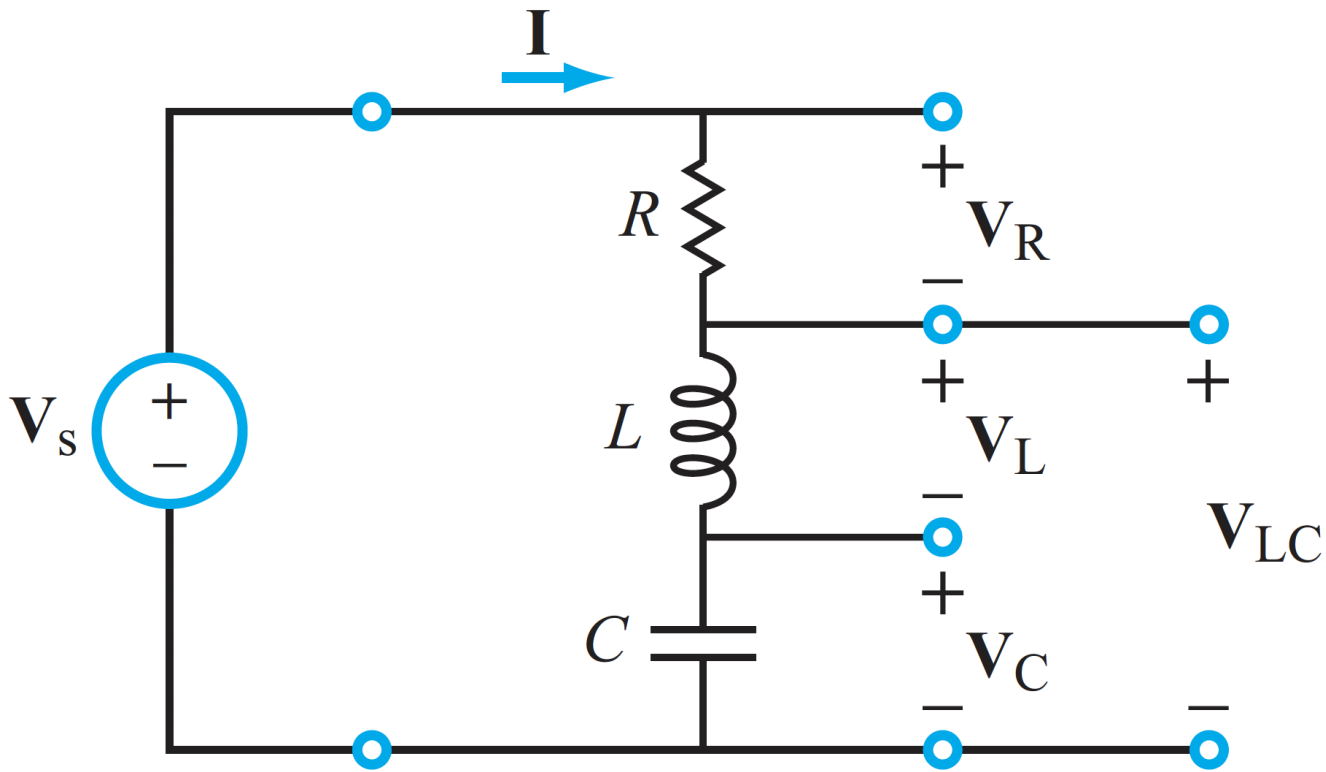
LASER = LIGHT Amplification by Stimulated Emission of Radiation



- Light bounces between two mirrors separated an integer number of wavelengths apart L .
- It gets amplified as it passes through the gain medium.
- Only the resonant wavelengths get amplified and emerge from the right as the laser light.



Bandpass Filter: Let's Get Detailed



**PASSIVE
BAND PASS FILTER**

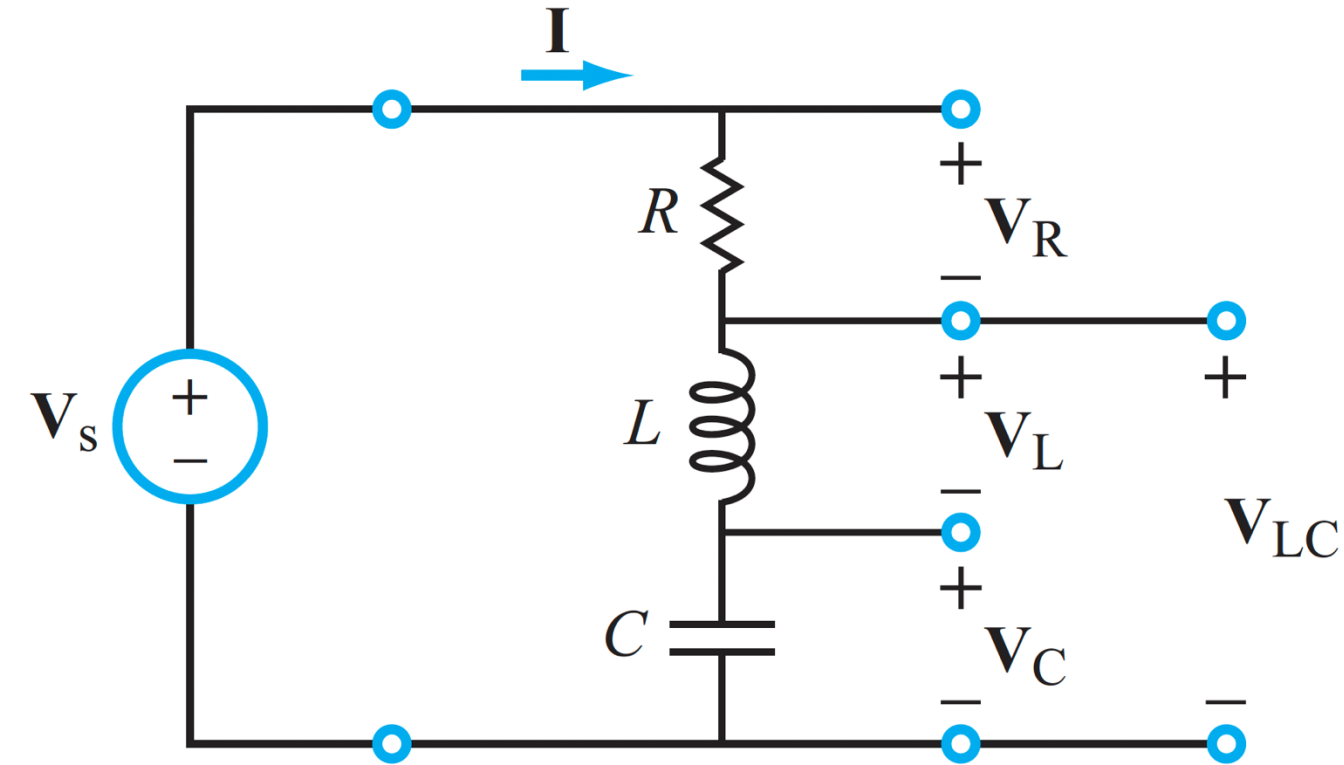
$$\hat{\mathbf{H}}_R(\omega) = \frac{\mathbf{V}_R}{\mathbf{V}_S} = \frac{j\omega RC}{j\omega RC + (1 - \omega^2 LC)}$$

$$|\hat{\mathbf{H}}_{BP}(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

$$\phi_{BP}(\omega) = \frac{\pi}{2} - \tan^{-1} \left[\frac{\omega RC}{1 - \omega^2 LC} \right]$$



Bandpass Filter: Input Impedance



**PASSIVE
BAND PASS FILTER**

$$Z_{in2} = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

The imaginary part becomes zero at a specific frequency ω_0

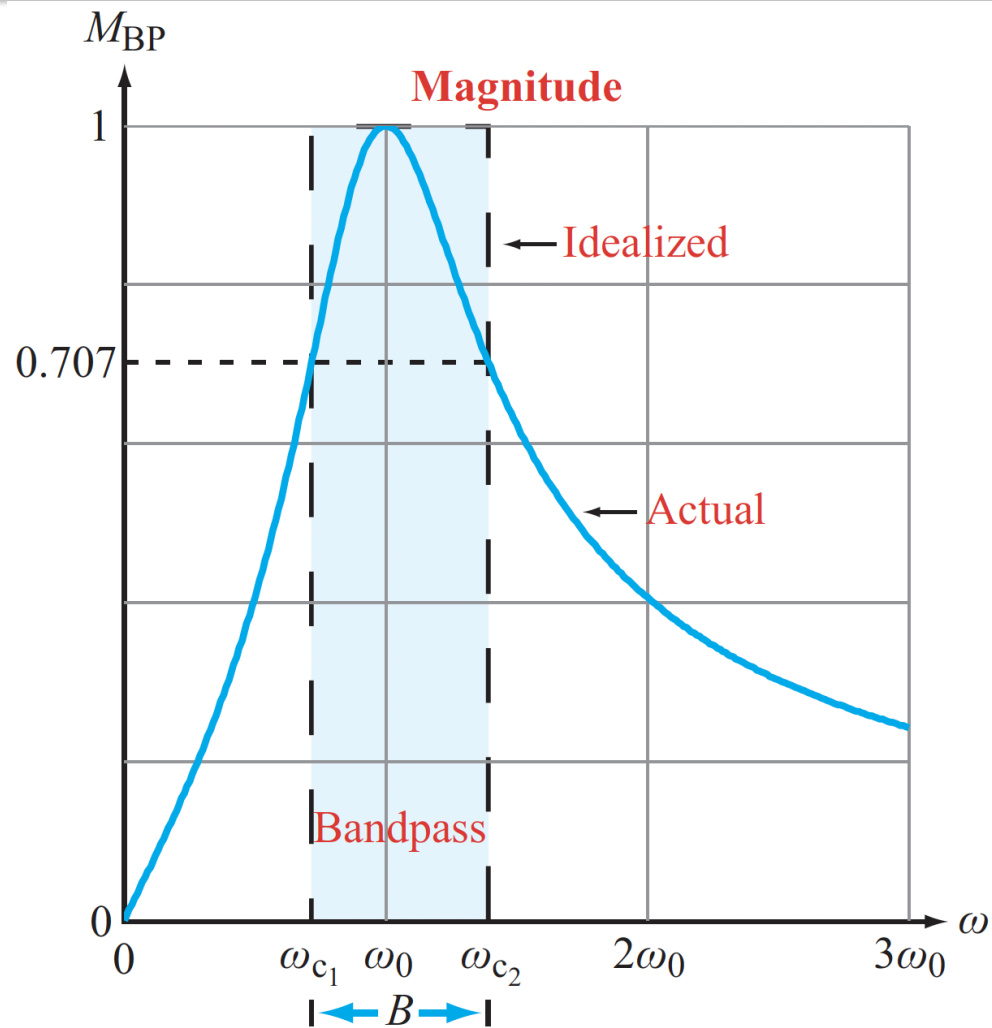
$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

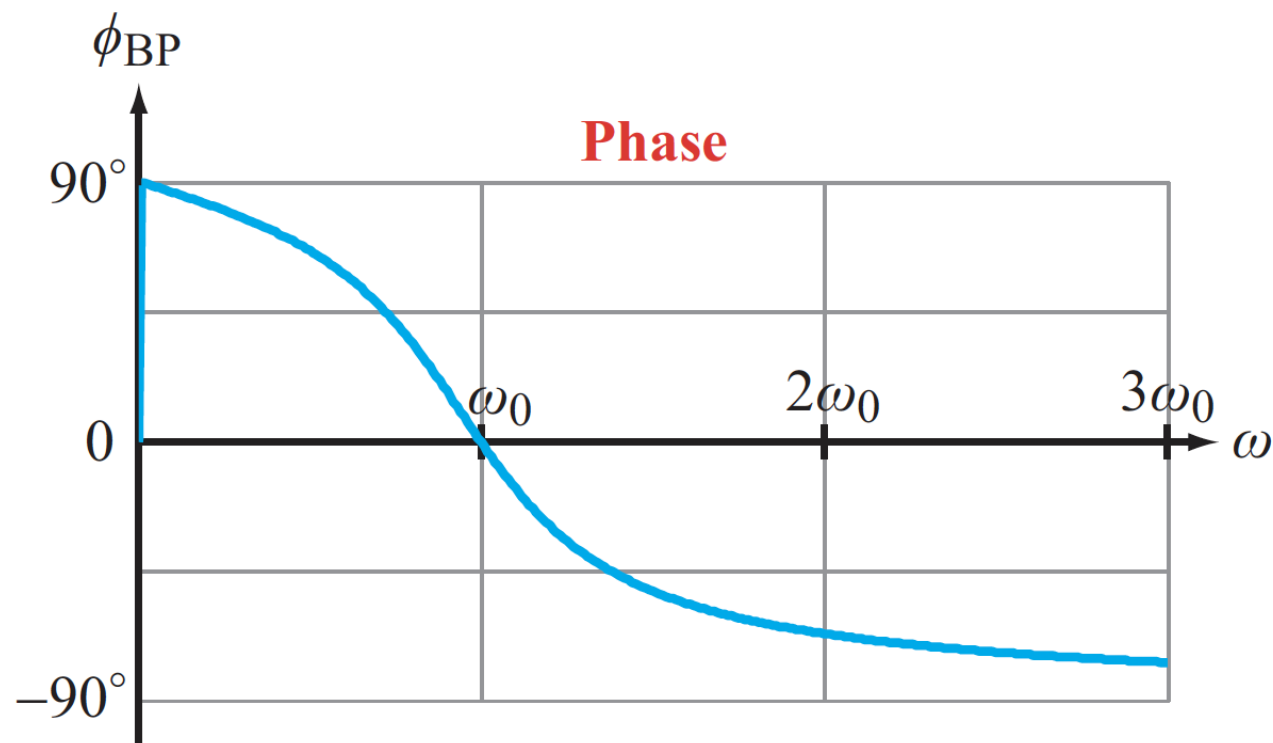
RESONANT FREQUENCY



Bandpass Response



$$|\hat{\mathbf{H}}_{BP}(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$



$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\phi_{BP}(\omega) = \frac{\pi}{2} - \tan^{-1} \left[\frac{\omega RC}{1 - \omega^2 LC} \right]$$



Calculating the Filter Bandwidth

$$\left| \hat{\mathbf{H}}_{BP}(\omega) \right| = M_{BP}(\omega) = \frac{\omega RC}{\sqrt{(\omega RC)^2 + (1 - \omega^2 LC)^2}}$$

Maximum value at $\omega_0 = 1$

Half power $\square M_{BP}(\omega) = \frac{1}{\sqrt{2}}$

$$\frac{(\omega RC)^2}{(\omega RC)^2 + (1 - \omega^2 LC)^2} = \frac{1}{2}$$

$$\square (\omega RC)^2 = (1 - \omega^2 LC)^2$$

$$\square \pm \omega RC = 1 - \omega^2 LC$$

$$\omega^2 LC \pm \omega RC - 1 = 0$$



Calculating the Filter Bandwidth

$$\omega^2 LC \pm \omega RC - 1 = 0$$

$$\text{Roots: } \mp \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_{c1} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_{c2} = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_{c2} - \omega_{c1} = \frac{R}{L}$$

BANDWIDTH

$$\omega^2 \pm \frac{R}{L} \omega - \frac{1}{LC} = 0$$

**OF THE 4 POSSIBLE ROOTS, KEEP
ONLY THE POSITIVE ONES**



Measuring the Quality of a Filter

We use the following dimensionless ratio Q to measure the quality (selectivity) of a filter

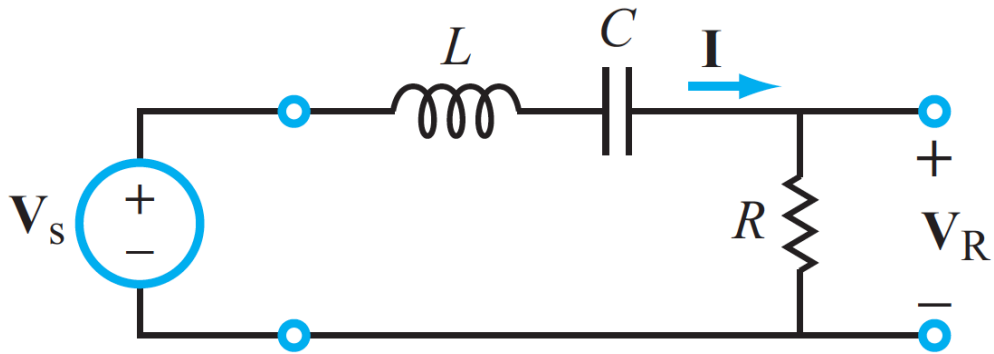
$$Q = \frac{\omega_0}{B} = \frac{1/\sqrt{LC}}{R/L} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

- For a filter to be considered more selective, should the Q value be low or high?
- How does Q change with resistance?
- What is the motivation to define such a number?

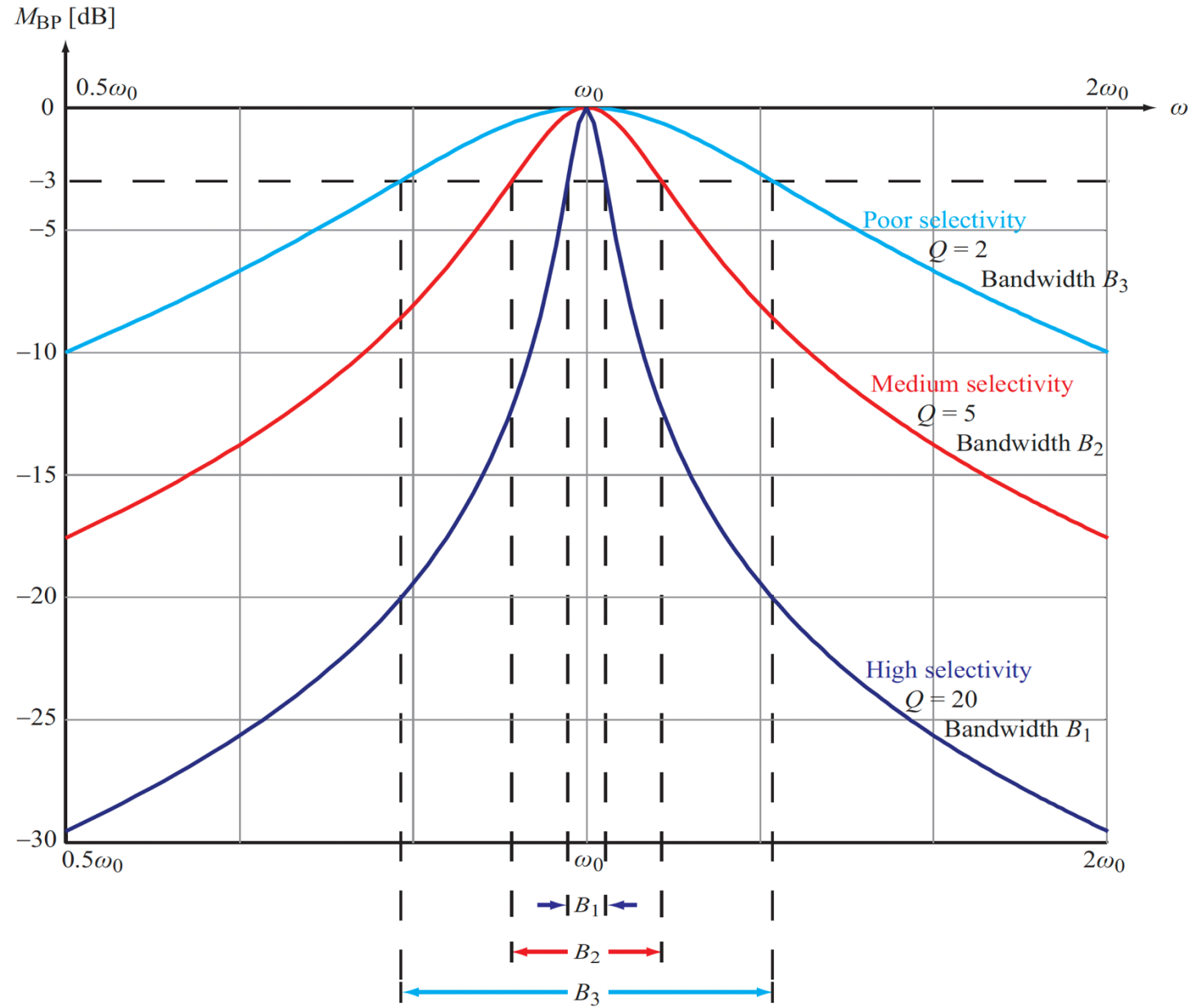


Quality Factor Visualization

RLC Bandpass



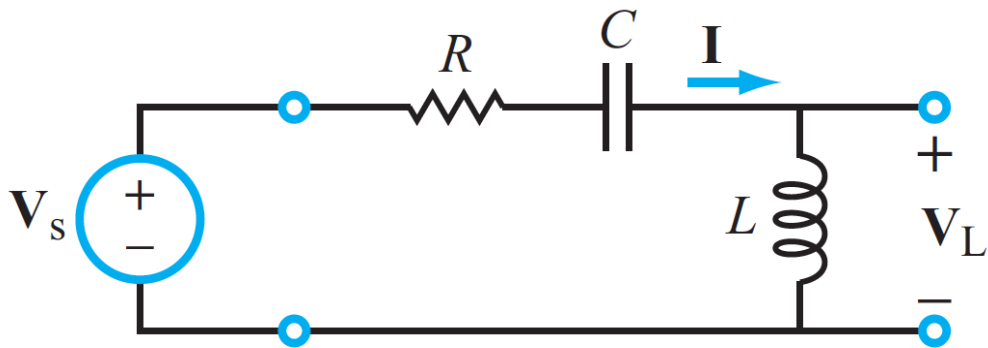
Voltage Divider Thinking:
L-C has zero impedance at
resonance frequency



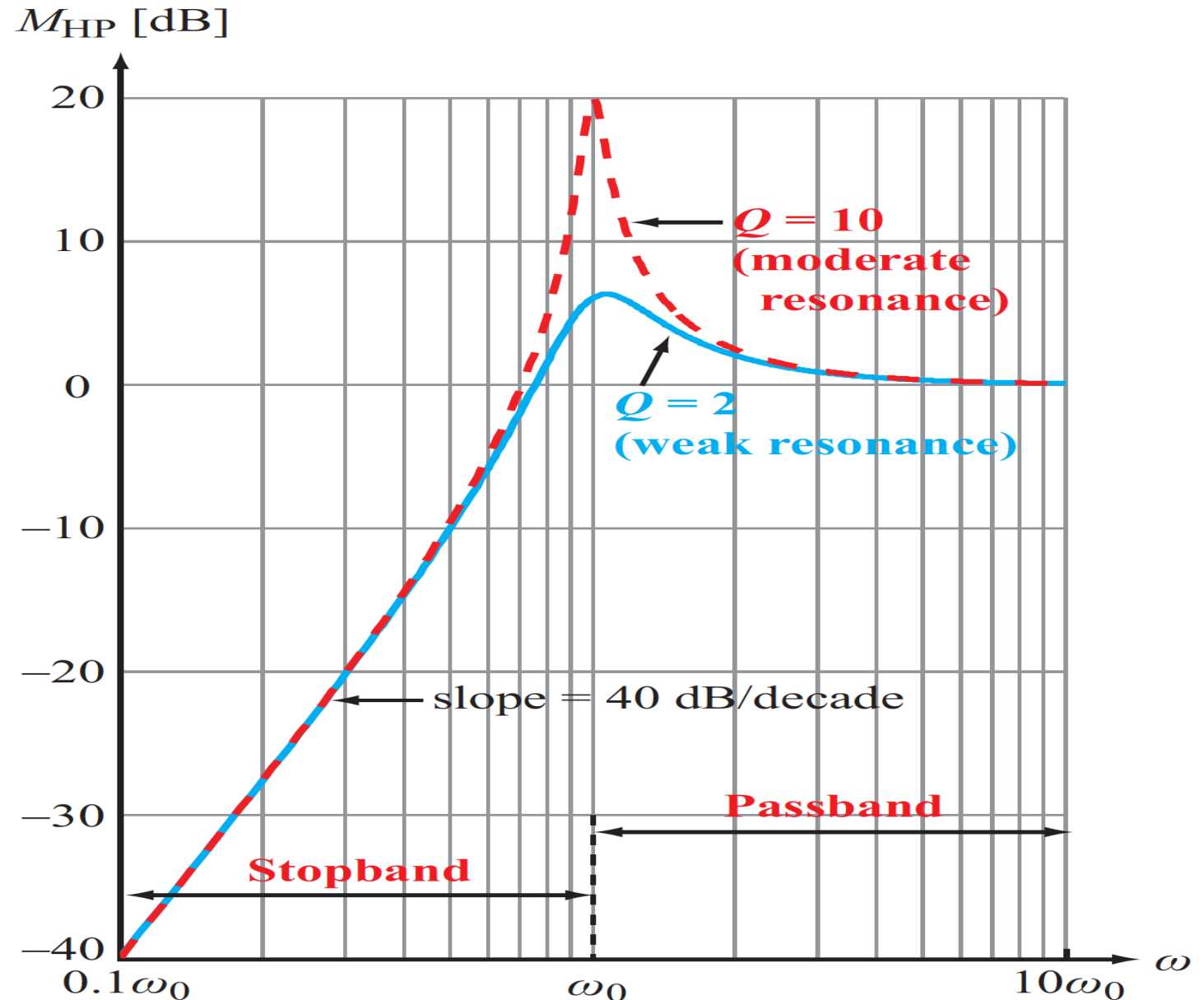


Quality Factor Example: RLC High-pass

RLC High-pass



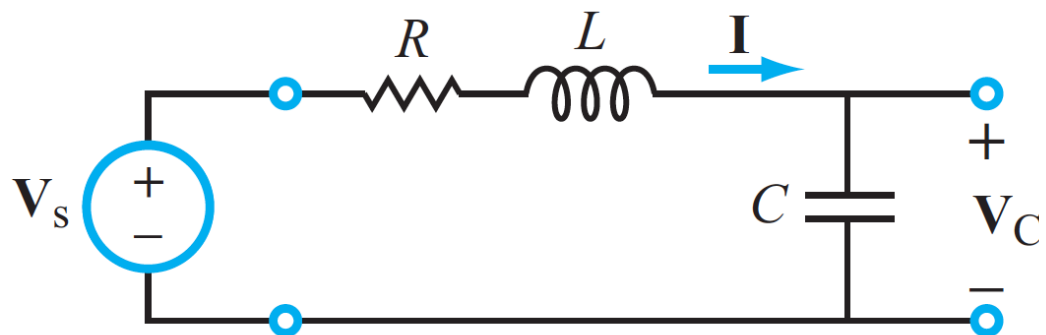
Voltage Divider Thinking:
 L has infinite impedance at high frequencies



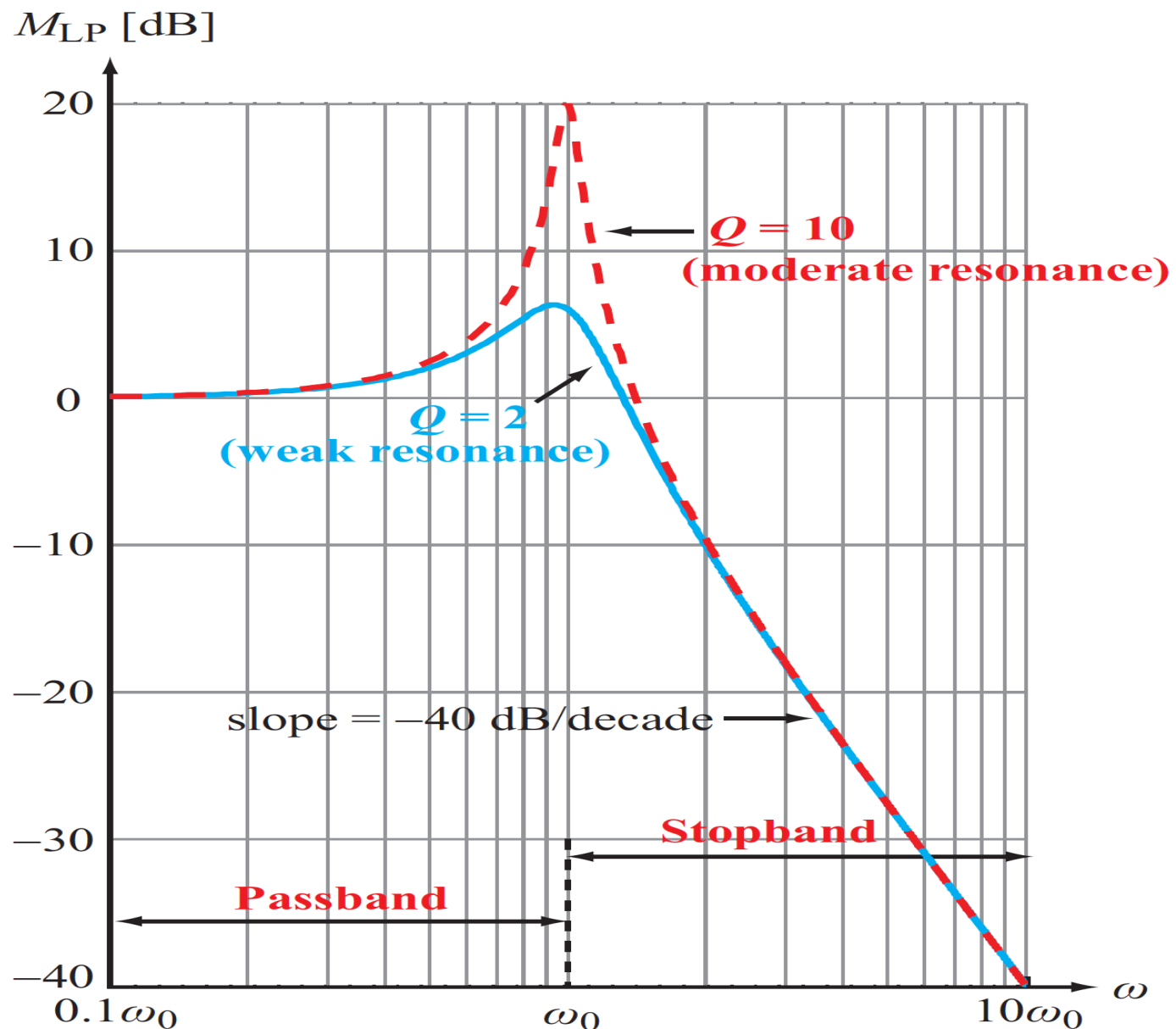


Quality Factor Example: RLC Low-pass

RLC Low-pass

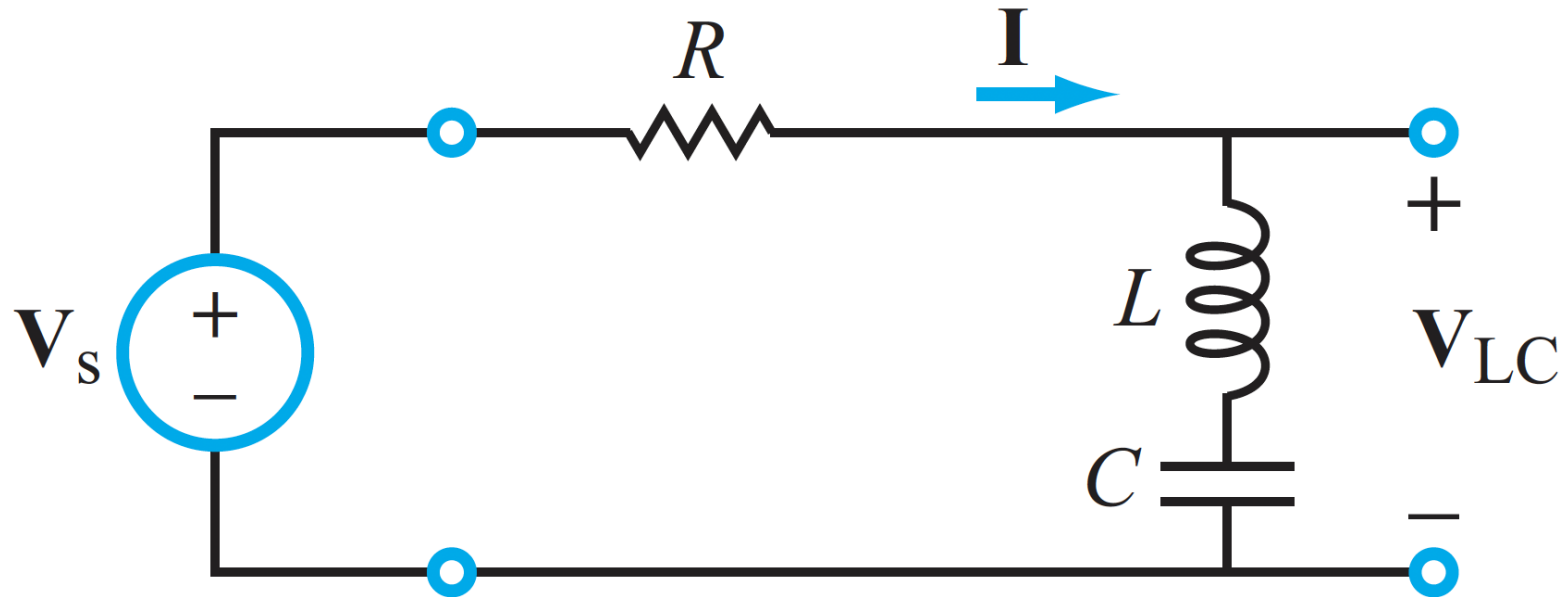


Voltage Divider Thinking:
C has zero impedance at high frequencies





Band Reject Filter

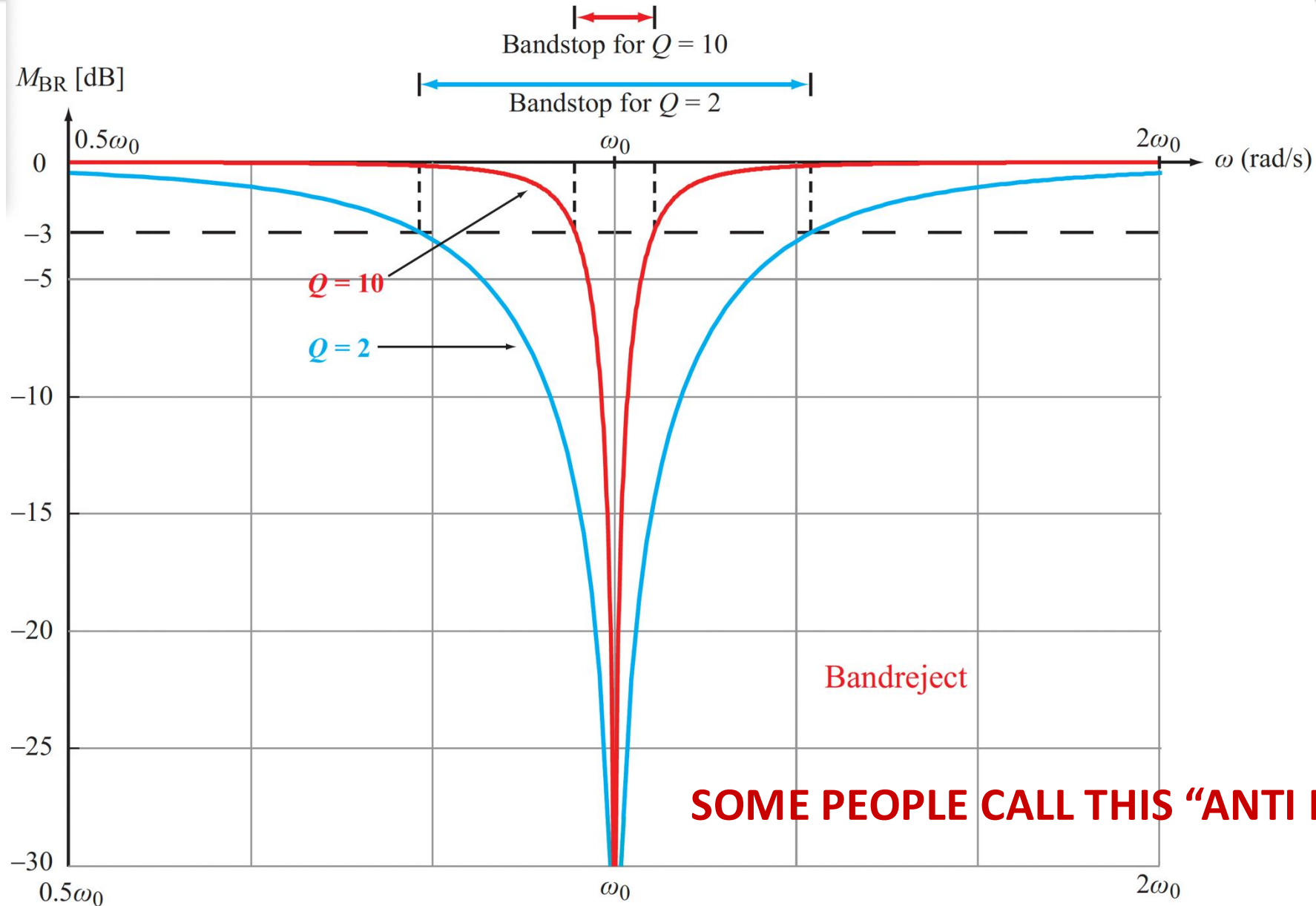


$$\hat{\mathbf{H}}_{BR}(\omega) = 1 - \hat{\mathbf{H}}_{BP}(\omega)$$

INTUITION: AT ω_0 , THE L - C COMBINATION HAS ZERO IMPEDANCE

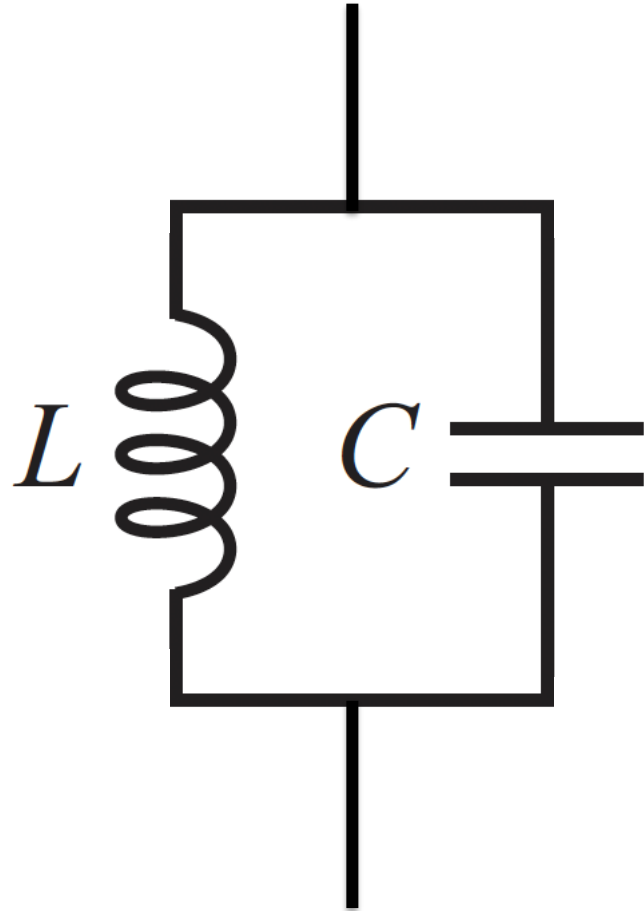


Band Reject Filter





Another Common Configuration



$$Z = Z_L || Z_C = \frac{Z_L Z_C}{Z_L + Z_C}$$

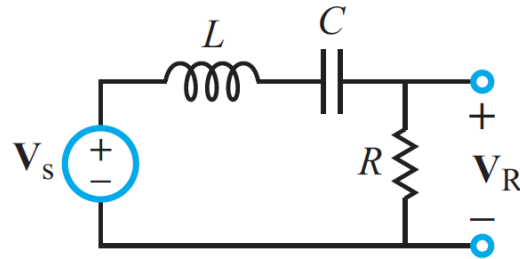
$Z \rightarrow \infty$ as $\omega \rightarrow \omega_0$ since

$$Z_L + Z_C \rightarrow 0$$



Summary of RLC Circuit Properties

RLC circuit



Transfer function

$$\mathbf{H} = \frac{\mathbf{V}_R}{\mathbf{V}_s}$$

Resonant frequency, ω_0

$$\frac{1}{\sqrt{LC}}$$

Bandwidth, B

$$\frac{R}{L}$$

Quality factor, Q

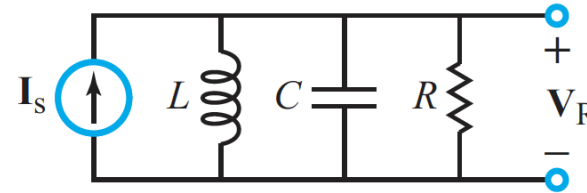
$$\frac{\omega_0}{B} = \frac{\omega_0 L}{R}$$

Lower half-power frequency, ω_{c1}

$$\left[-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

Upper half-power frequency, ω_{c2}

$$\left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$



CURRENT DIVIDER

$$\mathbf{H} = \frac{\mathbf{V}_R}{\mathbf{I}_s}$$

$$\frac{1}{\sqrt{LC}}$$

$$\frac{1}{RC}$$

$$\frac{\omega_0}{B} = \frac{R}{\omega_0 L}$$

$$\left[-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

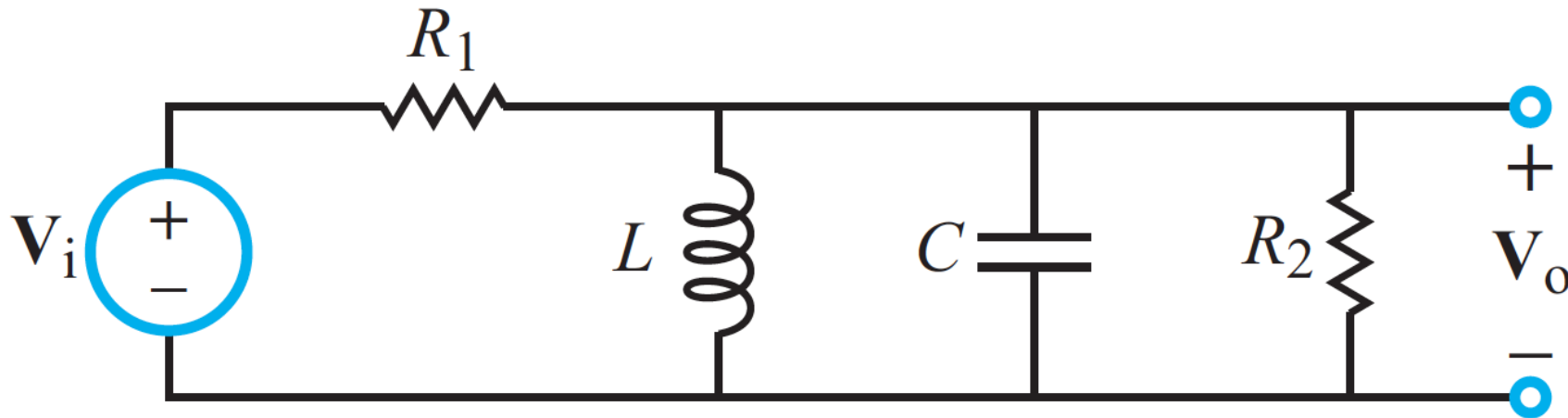
$$\left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

Notes: (1) The expression for Q of the series RLC circuit is the inverse of that for Q of the parallel circuit. (2) For $Q \geq 10$, $\omega_{c1} \approx \omega_0 - \frac{B}{2}$, $\omega_{c2} \approx \omega_0 + \frac{B}{2}$.



Activity

Derive the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o(\omega)/\mathbf{V}_i(\omega)$ for the following circuit, and the frequency ω_0 at which $\mathbf{H}(\omega)$ is purely real.



Correct Answers:

$$\frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{jR_2\omega L}{R_1R_2(1 - \omega^2 LC) + j\omega L(R_1 + R_2)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



Summary

Today, we learned more about signal filtering

- The basic filters include low-pass, high-pass, band-pass, and band-reject
- We discussed the powerful phenomenon of resonance
 - Happens when the input signal is matched to the (barely stable) system's natural frequency of oscillation
 - Can be a helpful thing, or a risk
- Bode plots are simple enough to be sketched by hand
 - We did not have enough time to discuss the quadratic terms, but the reasoning for them is similar



For the Next Class

- **Work through the textbook examples in Section 6-2**
- ***Study Section 6-4, 6-5, and 6-13 (Sampling Theorem).***
 - *This topic will serve as the bridge to your Digital Signal Processing course*
- **This is important:**
 - We want you to come prepared for each class by reading ahead, and have an idea of what to expect
 - Write down your questions and bring them with you
 - This is good for you
 - Advance reading will help you grasp the subject much more **easily**